Comprehensive Robustness Analysis of GCM, CCM, and OCB3*

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Abstract. Clarifying the robustness of authenticated encryption (AE) schemes, such as security under nonce misuse or Release of Unverified Plaintext (RUP), has significant importance due to the extensive use of AEs in real-world applications. We present a comprehensive analysis of the robustness of well-known standards, namely GCM, CCM, and OCB3. Despite many existing studies, we uncovered several robustness properties for them that were not known in the literature. In particular, we show that both GCM and CCM maintain authenticity under RUP. Moreover, CCM keeps this feature even if a nonce is misused. We show that this robust security is maintained if one block cipher call in CCM is omitted, resulting in an efficiency improvement of CCM since the proposal. Together with existing analysis, our work gives a complete picture of the robustness of these standards for the first time. Our results also imply several new robust AE schemes based on GCM and CCM.

Keywords: Authenticated Encryption · Robustness · GCM · CCM · OCB3

1 Introduction

Authenticated encryption (AE) [19, 69][4] is a type of symmetric-key encryption that simultaneously ensures the confidentiality and authenticity of messages. AE is widely acknowledged as a fundamental tool in practical cryptography and is widely deployed in practice, such as TLS for the Internet and WPA for Wifi access. A wide class of AE is *nonce-based*; that is, it takes a nonce as input, a value that never repeats, in encryption. The most popular and widely accepted prerequisite for nonce-based AE is *nonce-respecting*; the assumption that a nonce is non-repeating is crucial as a repeat of the nonce implies an immediate break of IND-CPA, the standard cryptographic notion for privacy/confidentiality. Another common prerequisite for AE is that the decryption routine must return a single error message when verification fails so that any information on the "unverified plaintext" does not leak.

* This is the full version of the original work published in CT-RSA 2025.

[4] AE with associated data (AEAD) is also a common name of our object of study.

However, in practice, these two prerequisites are sometimes violated, *i.e.*, AE can be *misused*. Misuse can happen due to various reasons, say poor randomness for nonce generation, misconfiguration of protocol, or side-channel/fault attacks, and has a devastating impact on real-world applications, such as [26, 66, 76]. Studying the impact of violation of these prerequisites, *i.e.*, *robustness*, is thus quite essential and practically relevant. A well-known example is Joux's "forbidden" attacks against GCM [48], which shows that even a single nonce repeat allows a universal forgery attack. Padding oracle attack [77] is another famous example regarding the second prerequisite. After these examples, AE's robustness has been a central topic in the literature. Violation of the first prerequisite is called Nonce Misuse (NM), and that of the second is called Release of Unverified Plaintext (RUP) [14] (as a special case of [27]), and studying the robustness of a given scheme means studying security under NM or RUP or both. A number of formal security notions capturing robustness have been proposed and studied [14, 16, 74]. The robustness of GCM and other popular schemes is a frequent question raised by implementors/users [8, 9, 10] and received attention in IETF (see *e.g.*, Bozhko [28]). Robustness of modern AE schemes, such as the candidates to CAESAR competition [2] and NIST Lightweight Cryptography (LWC) [4], has been extensively studied [3, 14, 45, 59, 75, 78]. In particular, NIST IR 8454 [3] mentioned that the robustness of the LWC finalists was considered for the final selection, together with a very detailed literature survey on the finalists, including their robustness analysis. The LWC winner Ascon also has received attention on its robustness [1, 18, 51].

Given this situation, it would be interesting to ask: do we fully understand the robustness of well-known standards, namely GCM [5], CCM [6], and OCB3 [50]? At first glance, it seems so since there are many existing studies (see Table 1 and Related Work), including the aforementioned forbidden attack by Joux. However, we reveal several robustness properties for them that are unknown (nor mentioned) in the literature. The purpose of this work is to report them in a comprehensive manner[5]. For the first time, we present the full picture of these standards' robustness together with existing analysis.

GCM and CCM are the two NIST AE standards (NIST SP800-38D [5] and SP800-38C [6]). They have been extensively used for TLS, IPSec, SSH, etc. CCM is the first specified AE mode for Wifi since WPA2 and is still mandated in the latest WPA3. It is also used for many resource-constrained networks, including LoRaWan, Bluetooth, and Zigbee, to name a few. See Rogaway's CRYPTREC report [72] for more information on their real-world applications. OCB3 is the third (and the final) version of the seminal OCB mode family and is an Internet standard (RFC 7253) [7].

For GCM, we show that it maintains authenticity even if it releases unverified plaintext, also known as INTegrity under RUP (INT-RUP) [14], irrespective of nonce length. Except for Ashur *et al.*'s result [16] (ADL17) on nonce misuse resilience privacy (see Table 1), GCM is considered to lack the robustness of any

[5] We do not consider leakage-resistance [22, 23, 64] as it requires additional security assumptions on side-channel attacks for implementations of (specific) components.

known kind. Our result shows that this is not the case, and hence, we think this is something unexpected.

For CCM, we show an even stronger result: it maintains INT-RUP even when nonce may repeat in encryption, which we refer to as Nonce-Misuse-Resistant (NMR) INT-RUP. This notion gives the strongest authenticity property that we consider in this paper. In addition, we show that CCM maintains privacy in the sense of Nonce Misuse resiLience (NML), a relaxed notion from nonce misuse resistance introduced by ADL17. These robustness properties are realized by CCM's prefix-free encoding applied to the internal CBC-MAC. The prefix-freeness was also the key of the existing nonce-respecting proofs [36, 47, 78], however, never used for proving robustness. These results indicate that CCM is much more robust than GCM (and OCB3, see Table 1). Despite its wide real-world adoption, CCM has received surprisingly less attention than GCM or OCB3 from a provable security perspective after Jonsson's initial proof [47]. Our results on CCM is connected to the existing results, say Petrank and Rackoff [65] for CBC-MAC taking prefix-free inputs. However, the structure of CCM is more involved which requires us to conduct careful dedicated analysis instead of a black-box use of existing results including Petrank and Rackoff. Our results could benefit protocols/applications adopting CCM to provide an in-depth defense.

For OCB3, the previous analysis covers most of the existing robustness notions. We present a study on nonce misuse resilience. Specifically, we show OCB3's privacy in the sense of nonce misuse resilience. Authenticity is broken in a general case, however, we point out that under a restriction on associated data, authenticity is also maintained. These results compensate for the original analysis of ADL17, which mainly focuses on the first version of OCB.

Table 1 shows the summary of our results together with existing results. For comparison, we also list the results for ChaCha20-Poly1305 [61], where all the robustness properties we consider are already known in the literature.

Benefits of Our Results. Our results have several implications on AE constructions. First, given that GCM and CCM are INT-RUP secure, we can convert them into a RUP-secure AE [14], a secure AE for both privacy and authenticity under RUP, in an efficient (near) black-box way, thanks to a method by Andreeva *et al.* [14]. An earlier RUP-secure AE based on GCM (GCM-RUP [16]) needs to modify the algorithm. No CCM-based scheme is known. RUP-secure AE is relevant as a countermeasure against real-world attacks such as Efail [66]. Second, our security proofs reveal that one block cipher call in CCM can be omitted, maintaining all the security properties, including those we prove in this paper. We call this optimized variant CCM2. Surprisingly, we found such an optimization after more than 15 years of being standardized by NIST. This improvement is particularly effective when messages are short [13]. Note that short inputs are common for wireless low-power communication protocols and even for Internet protocols, where CCM is often used.

Related Work. Provable security of GCM under the nonce-respecting scenario has been extensively studied [21, 41, 46, 54, 55, 57, 60, 62]. Design issues such as

Table 1: Summary of results. GCM is NML-Priv secure only if nonce is 96 bits and broken otherwise. OCB3 is NML-Auth secure only if AD is fixed. CCM2 is our optimized version of CCM.

	RUP		NMR		NML		RUP+NMR
	Priv (PA1)	INT-RUP	Priv	Auth	Priv	Auth	Auth
GCM	✗ [14]	✓ Sect. 4	✗ [Classical]	✗ [48]	✓/✗ [16]	✗ [16]	✗ [48]
CCM/CCM2	✗ [14]	✓ Sect. 5	✗ [Classical]	✓ Sect. 5	✓ Sect. 6	✓ Sect. 5	✓ Sect. 5
OCB3	✗ [14]	✗ [14]	✗ [Classical]	✗ [78]	✓ Sect. 7	✓/✗ Sect. 7, [78]	✗ [78]
ChaChaPoly	✗ [14]	✓ [42, 43]	✗ [Classical]	✗ [49]	✓ [16]	✓ [16]	✗ [49]

security degradation for short tag or weak keys have been discussed [11, 34, 38, 56, 67]. Dodis *et al.* studied key committing security of GCM [32].

Provable security of CCM under the nonce-respecting scenario was proved by Jonsson [47]. Rogaway and Wagner [68] reviewed the design and pointed out several deficiencies. Fouque *et al.* [36] presented a security bound for a variant of CCM and forgery attacks on (a general form of) CCM based on nonce repeat. Their nonce-misuse forgery attack against the original CCM has birthday complexity and hence does not contradict our result. Gjiriti *et al.* [37] showed provable security of a variant of CCM with variable tag stretch. Menda *et al.* [58] showed key committing attack against CCM. Very recently, Zhang *et al.* [79] proved its multi-user security.

Provable security of OCB3 under the nonce-respecting scenario has been proved by Krovetz and Rogaway [50]. The bound was improved by Bhaumik and Nandi [25]. An authenticity attack under RUP was shown by [14]. Liénardy and Lafitte [52] pointed out an issue of the current specification on the valid nonce length.

Vaudenay and Vizár [78] showed a nonce-misuse analysis on CAESAR candidates and GCM, CCM and OCB3. Rogaway and Shrimpton [74] introduced the notion of nonce-misuse resistance and proposed SIV, an offline AE with nonce-misuse resistance. Robust AE (RAE) [39] is a class of (offline) AEs having full protection against nonce misuse and RUP. Online AE [35] is a class of nonce-based AE that has partial privacy protection against nonce misuse while online processing. Hoang *et al.* [40] discussed its practical relevance and limitation.

2 Preliminaries

2.1 Basic notations

For non-negative integers i and j with $i < j$, let $[i..j] \coloneqq \{i, i+1, i+2, \ldots, j\}$. For any list $X = (X[1], \ldots, X[x])$ and $i, j \in [1..x]$, $i < j$, we write $X[i..j]$ to mean $(X[i], \ldots, X[j])$. Let $\{0,1\}^*$ be the set of all bit strings, including the empty string ε, the single element in $\{0,1\}^0$. For $X \in \{0,1\}^*$, $|X|$ denotes its bit length and $|X|_n$ denotes $\lceil |X|/n \rceil$. We define the parsing of a string into n-bit blocks for a positive integer n by $(X[1], X[2], \ldots, X[m]) \xleftarrow{n} X$, where $X[1] \parallel X[2] \parallel \ldots \parallel X[m] = X$, $|X[i]| = n$ for $1 \leq i < m$ and $0 < |X[m]| \leq n$ when $|X| > 0$. When $|X| = 0$ (*i.e.*, $X = \varepsilon$), we let $X[1] \xleftarrow{n} X$ with $X[1] = \varepsilon$. A concatenation of bit strings X and Y is denoted as $X \parallel Y$, or XY, in case no confusion is possible. Let 0^i be the string of i zero bits. Hence, we write 10^i for $1 \parallel 0^i$. For $X \in \{0,1\}^*$ with $|X| \geq i$, $\mathtt{msb}_i(X)$ is the first (left) i bits of X, and $\mathtt{lsb}_i(X)$ is the last (right) i bits of X. Let $\langle X \rangle_c$ be the c-bit standard representation of $X \in [0..2^c - 1]$. If X is uniformly chosen from the set $\mathcal{X}$, we write $X \xleftarrow{\$} \mathcal{X}$. For any $X \in \{0,1\}^*$ and a positive integer n, let $\mathsf{pad}_n(X)$ be $X \parallel 00\ldots0$ so that $|\mathsf{pad}_n(X)|$ be the minimum number of multiple of n. If $|X|$ is a positive multiple of n, $\mathsf{pad}_n(X) = X$, and $\mathsf{pad}_n(\varepsilon) = \varepsilon$. Note that pad_n is not injective.

Oracles and advantage. Suppose $\mathcal{A}$ is an adversary in a game. $\mathcal{A}$ can query to s oracles, $\mathcal{O}_1, \ldots, \mathcal{O}_s$, in any order. In the case of a distinguishing game, $\mathcal{A}$ outputs a bit after the queries, which is a random variable whose probability space is defined under $\mathcal{A}$ and $\{\mathcal{O}_i\}_{i \in [1..s]}$. Let $[\mathcal{A}^{\mathcal{O}_1, \ldots, \mathcal{O}_s} = 1]$ denote the event that this bit is 1. We call $\mathcal{O}_i$ the i-th oracle in the game. By writing $\mathbf{Adv}_{\Pi}^{\mathtt{xxx}}(\mathcal{A})$, we mean *advantage* of the adversary $\mathcal{A}$ in the game $\mathtt{xxx}$ involving Π. We say Π is $\mathtt{xxx}$-secure if the corresponding advantage is negligible for all computationally bounded adversaries.

(Tweakable) Block cipher and random primitives. Any keyed function $F : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with key space $\mathcal{K}$, we may write $F_K(x)$ to denote $F(K, x)$. Unless otherwise specified, we assume $K \xleftarrow{\$} \mathcal{K}$ in evaluation of $F_K(x)$. A block cipher $E : \mathcal{K} \times \mathcal{M} \to \mathcal{M}$ is a keyed function such that for any $K \in \mathcal{K}$, $E(K, \cdot)$ is a permutation over the message space $\mathcal{M}$. We write $E_K^{-1}(\cdot)$ to denote the decryption function. A tweakable block cipher (TBC) [53] $\widetilde{E} : \mathcal{K} \times \mathcal{TW} \times \mathcal{M} \to \mathcal{M}$ is a keyed function such that for any $K \in \mathcal{K}$ and *tweak* $W \in \mathcal{TW}$, $\widetilde{E}(K, W, \cdot)$ is a permutation over $\mathcal{M}$. Tweak is typically used as a public/chosen value in contrast to the secret key. Let $\mathrm{Perm}(n)$ be the set of all permutations of n bits, and $\mathrm{Func}(n, m)$ be the set of all functions: $\{0,1\}^n \to \{0,1\}^m$. A uniform random permutation (URP) $\mathsf{P} : \{0,1\}^n \to \{0,1\}^n$ is a random permutation uniformly chosen over $\mathrm{Perm}(n)$. A uniform random function (URF) $\mathsf{R} : \{0,1\}^n \to \{0,1\}^m$ is a random function uniformly chosen over $\mathrm{Func}(n, m)$. A tweakable uniform random permutation (TURP) $\widetilde{\mathsf{P}} : \mathcal{TW} \times \mathcal{M} \to \mathcal{M}$ consists of $|\mathcal{TW}|$ independent instances of P over $\mathcal{M}$ and the first argument (tweak) specifies the URP used.The security of TBC $\widetilde{E}$ is defined as the following distinguishing probability.

$$\mathbf{Adv}_{\widetilde{E}}^{\mathtt{TPRP}}(\mathcal{A}) := |\Pr[\mathcal{A}^{\widetilde{E}_K} = 1] - \Pr[\mathcal{A}^{\widetilde{\mathsf{P}}} = 1]|,$$

where $\widetilde{\mathsf{P}}$ has the same domain and range as $\widetilde{E}_K$. When $\mathcal{TW}$ is a singleton, the security of block cipher E_K is obtained by the above security, denoted by $\mathbf{Adv}_{E}^{\mathtt{PRP}}(\mathcal{A})$.

H-Coefficient. All our proofs use the standard H-Coefficient technique [31,63]. See App. A for its basics.

2.2 Authenticated encryption

A nonce-based AE scheme $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ is defined over a key space $\mathcal{K}$, a nonce space $\mathcal{N}$, an associated data (AD) space $\mathcal{AD}$, a message space $\mathcal{M}$, and a tag space $\mathcal{T} = \{0,1\}^\tau$ for some fixed tag length τ. We use ν throughout the paper to denote the nonce length in bits, hence $\mathcal{N} = \{0,1\}^\nu$. An AD is information that is not to be kept confidential, but its integrity must be ensured. For example, network encryption will treat the protocol header as an AD. Formally, AE is a tuple of an encryption function $\Pi.\mathcal{E} \colon \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \to \mathcal{M} \times \mathcal{T}$, and a decryption function $\Pi.\mathcal{D} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \times \mathcal{T} \to \mathcal{M} \cup \{\perp\}$, where symbol $\perp$ indicates a verification failure. We require $\Pi.\mathcal{D}_K(N, A, C, T) = M$ if $\Pi.\mathcal{E}_K(N, A, M) = (C, T)$ for soundness. Moreover, we assume $|C| = |M|$; all our target schemes meet this condition. Basically, each nonce in encryption must be unique, which we call *nonce-respecting* (NR) model. Note that a nonce in decryption has no restrictions; it may repeat and it may collide with one used in encryption. A nonce-respecting adversary (NR adversary) may choose nonce arbitrarily in encryption, except that this condition must be respected [71]. Let $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ be an AE scheme. Let \$ be the random-bit oracle that takes (N, A, M) and returns random $|M| + \tau$ bits. The standard security notions of Π against NR adversaries are Priv (confidentiality/privacy) and Auth (authenticity/integrity), defined below.

Definition 1 (Priv and Auth). *Let $\mathcal{A}$ be a* Priv *adversary against* Π *and let $\mathcal{B}$ be an* Auth *adversary against* Π*.*

$$\mathbf{Adv}_{\Pi}^{\texttt{Priv}}(\mathcal{A}) \coloneqq |\Pr[\mathcal{A}^{\Pi.\mathcal{E}_K} = 1] - \Pr[\mathcal{A}^{\$} = 1]|,$$
$$\mathbf{Adv}_{\Pi}^{\texttt{Auth}}(\mathcal{B}) \coloneqq \Pr[\mathcal{B}^{\Pi.\mathcal{E}_K, \Pi.\mathcal{D}_K} \text{ forges}],$$

where $\mathcal{A}$ and $\mathcal{B}$ are nonce-respecting. The event $[\mathcal{B}^{\Pi.\mathcal{E}_K, \Pi.\mathcal{D}_K}$ forges$]$ means that $\mathcal{B}$ receives some $M' \neq \perp$ from $\Pi.\mathcal{D}_K$. $\mathcal{B}$ does not forward queries from $\Pi.\mathcal{E}_K$ to $\Pi.\mathcal{D}_K$, namely, $\mathcal{B}$ does not query (N, A, C, T) to $\Pi.\mathcal{D}_K$ after obtaining (N, A, M, C, T) as a result of query-response to $\Pi.\mathcal{E}_K$.

It is also possible to consider a combined notion that captures Priv and Auth. While this is convenient, since our purpose is a fine-grained analysis, we always treat privacy and authenticity notions separately. This applies to the various robustness notions described in the next section.

3 Robustness Notions of Authenticated Encryption

3.1 Security notions under nonce misuse

Rogaway and Shrimpton [74] introduced the concept of *Nonce-Misuse Resistance* (NMR), best-possible security notions under possibly repeating nonces in encryptions. NMR security notions are as follows.

Definition 2 (NMR-Priv and NMR-Auth [74]). *Let $\mathcal{A}$ be an NMR-Priv adversary against Π and let $\mathcal{B}$ be an NMR-Auth adversary against Π.*

$$\mathbf{Adv}_{\Pi}^{\texttt{NMR-Priv}}(\mathcal{A}) := |\Pr[\mathcal{A}^{\Pi.\mathcal{E}_K} = 1] - \Pr[\mathcal{A}^{\$} = 1]|,$$
$$\mathbf{Adv}_{\Pi}^{\texttt{NMR-Auth}}(\mathcal{B}) := \Pr[\mathcal{B}^{\Pi.\mathcal{E}_K, \Pi.\mathcal{D}_K} \text{ forges}],$$

where $\mathcal{A}$ and $\mathcal{B}$ may query to the first oracle with repeating nonces, but does not query a repeated tuple of (N, A, M). $\mathcal{B}$ does not forward queries from the first oracle to the second oracle.

We remark that NMR-Priv requires $\Pi.\mathcal{E}$ to be offline, *i.e.*, any ciphertext bit must reflect the whole encryption input. This limits applicability in particular for long messages. GCM and OCB3 are online schemes; it is inherently impossible to meet this notion. CCM is not online, but the encryption part is nevertheless a counter mode and fails to meet NMR-Priv [78].

Nonce-misuse resilience. Ashur *et al.* [16] proposed the concept of *Nonce-Misuse resiLience* (NML), which is a relaxed form of NMR. Intuitively, NML security notions require that the attacker must win the game by making a query with a nonce that is not misused (repeated). Conversely, an AE scheme is NML-secure if a repeat of nonce N does not threaten the security of encryption/decryption with a different nonce from N. While weaker than NMR, NML notions can be fulfilled by online schemes. If nonce is determined by (*e.g.*) recipient ID or timestamp, NML is practically relevant.

Definition 3 (NML-Priv and NML-Auth [16]). *Let $\mathcal{A}$ be an NML-Priv adversary against Π and let $\mathcal{B}$ be an NML-Auth adversary against Π.*

$$\mathbf{Adv}_{\Pi}^{\texttt{NML-Priv}}(\mathcal{A}) := |\Pr[\mathcal{A}^{\Pi.\mathcal{E}_K, \Pi.\mathcal{E}_K} = 1] - \Pr[\mathcal{A}^{\$, \Pi.\mathcal{E}_K} = 1]|,$$
$$\mathbf{Adv}_{\Pi}^{\texttt{NML-Auth}}(\mathcal{B}) := \Pr[\mathcal{B}^{\Pi.\mathcal{E}_K, \Pi.\mathcal{D}_K} \text{ forges}],$$

where $\mathcal{A}$ is nonce-respecting with respect to the first oracle, i.e., let $\mathcal{N}_1$ and $\mathcal{N}_2$ be the multisets of nonces used by the queries to the first (nonce-respecting) and the second (nonce-misusing) oracles; $\mathcal{N}_1 \cap \mathcal{N}_2 = \emptyset$ and each $N \in \mathcal{N}_1$ has a multiplicity one. $\mathcal{N}_2$ may contain nonces with multiplicity ≥ 2. Similarly for $\mathcal{B}$, we define $\mathcal{N}_1$ and $\mathcal{N}_2$. Both may contain nonces with multiplicity ≥ 2. However, any $N \in \mathcal{N}_1 \cap \mathcal{N}_2$ must have multiplicity one in $\mathcal{N}_1$. In addition, as with the standard Auth *notion, a trivial forgery is prohibited.*

3.2 Security notions under release of unverified plaintext

Release of Unverified Plaintext (RUP) was formalized by Andreeva *et al.* [14]. RUP is the game environment that leaks the unverified plaintext to the adversary at decryption independent of the verification result. To formalize RUP security games, we require that $\Pi.\mathcal{D}$ is decomposed into the unverified decryption function, $\Pi.u\mathcal{D} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \times \mathcal{T} \to \mathcal{M}$ and the verification function, $\Pi.\mathcal{V} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \times \mathcal{T} \to \{\top, \perp\}$. The normal decryption combines them as $\Pi.\mathcal{D}_K(N, A, C, T) = M$ if $\Pi.u\mathcal{D}_K(N, A, C, T) = M$ and $\Pi.\mathcal{V}_K(N, A, C, T) = \top$, and $\Pi.\mathcal{D}_K(N, A, C, T) = \perp$ if $\Pi.\mathcal{V}_K(N, A, C, T) = \perp$. That is, when $\Pi.u\mathcal{D}_K(N, A, C, T) = M$, M denotes the correct decrypted plaintext if the input tuple is authentic, and otherwise M is something that would be discarded by the decryption oracle in the classical authenticity notions. Most existing AE schemes, including our targets, fulfill the aforementioned assumption. We will write $\Pi = (\Pi.\mathcal{E}, \Pi.u\mathcal{D}, \Pi.\mathcal{V})$ when we discuss about RUP security of Π. As described, $\Pi.\mathcal{D}$ is uniquely determined from $\Pi.u\mathcal{D}$ and $\Pi.\mathcal{V}$.

Privacy notion under RUP is Plaintext Awareness (PA), and PA is classified into a basic notion (PA1) and a stronger notion (PA2). As they are irrelevant to our analysis, we omit the definitions here. The authenticity notion under RUP is called INT-RUP (for INTegrity under RUP) [14].

Definition 4 (INT-RUP). *Let $\mathcal{A}$ be the INT-RUP adversary against $\Pi = (\Pi.\mathcal{E}, \Pi.u\mathcal{D}, \Pi.\mathcal{V})$.*

$$\mathbf{Adv}_{\Pi}^{\mathtt{INT\text{-}RUP}}(\mathcal{A}) := \Pr[\mathcal{A}^{\Pi.\mathcal{E}_K, \Pi.u\mathcal{D}_K, \Pi.\mathcal{V}_K} \text{ forges}],$$

where $\mathcal{A}$ is nonce-respecting, i.e., never repeats nonce in queries to $\Pi.\mathcal{E}$. Moreover, $\mathcal{A}$ does not forward queries from $\Pi.\mathcal{E}$ to $\Pi.\mathcal{V}$. Queries to $\Pi.u\mathcal{D}$ have no restrictions.

We can combine INT-RUP with the nonce-misuse scenario, yielding a more robust authenticity notion.

Definition 5 (NMR-INT-RUP). *The nonce-misuse resistance INT-RUP (NMR-INT-RUP) advantage, $\mathbf{Adv}_{\Pi}^{\mathtt{NMR\text{-}INT\text{-}RUP}}(\mathcal{A})$, is defined in the same manner as INT-RUP, except that $\mathcal{A}$ is allowed to repeat nonces in queries to the first oracle.*

By definition, we have the following implications: (1) NMR-Priv $\to$ NML-Priv $\to$ Priv, and (2) NMR-Auth $\to$ NML-Auth $\to$ Auth, and (3) NMR-INT-RUP $\to$ INT-RUP, and (4) NMR-INT-RUP $\to$ NMR-Auth.

NMR-INT-RUP and related notions have been studied for on-line AE scheme [24], which is a class of AEs providing a relaxed NM security while enabling online processing [15]. We remark that none of our targets is an on-line AE.

4 Authenticity of GCM under RUP

We show that GCM is INT-RUP secure. As described in the introduction, we think this result is kind of unexpected since GCM is considered to lack robustness other than NML-Priv with a restriction to 96-bit nonce [16].

4.1 Specification of GCM

We briefly present the specification of GCM. Let $n = 128$ and E be an n-bit block cipher. For $X \in \{0,1\}^\ell$ such that $\langle X \rangle_\ell = x_{\ell-1} \cdots x_1 x_0$, let $\mathsf{int}(X)$ be an integer

Algorithm $\mathsf{GCM}.\mathcal{E}[E_K](N, A, M)$

1. $C \leftarrow \mathsf{GCTR}[E_K](N, C)$
2. $T \leftarrow \mathsf{GMAC}.\mathcal{T}[E_K](N, A, C)$
3. **return** (C, T)

Algorithm $\mathsf{GCM}.u\mathcal{D}[E_K](N, A, C, T)$

1. $M \leftarrow \mathsf{GCTR}[E_K](N, C)$
2. **return** M

Algorithm $\mathsf{GCM}.\mathcal{D}[E_K](N, A, C, T)$

1. $b = \mathsf{GCM}.\mathcal{V}[E_K](N, A, C, T)$
2. **if** $b = \bot$ **then return** $\bot$
3. **else** $M \leftarrow \mathsf{GCM}.u\mathcal{D}[E_K](N, A, C, T)$
4. **return** M

Algorithm $\mathsf{GCM}.\mathcal{V}[E_K](N, A, C, T)$

1. $\widehat{T} \leftarrow \mathsf{GMAC}.\mathcal{T}[E_K](N, A, C)$
2. **if** $T = \widehat{T}$ **then return** $\top$
3. **else return** $\bot$

Algorithm $\mathsf{GCTR}[E_K](N, M)$

1. $L \leftarrow E_K(0^n)$
2. **if** $\nu = 96$ **then** $\mathsf{Ctr0} \leftarrow N \,\|\, 0^{31}1$
3. **else then** $\mathsf{Ctr0} \leftarrow \mathsf{GHASH}_L(\varepsilon, N)$
4. $(M[1], \ldots, M[m]) \xleftarrow{n} M$
5. **for** $i = 1$ **to** m **do**
6. $\quad C[i] \leftarrow M[i] \oplus \mathtt{msb}_{|M[i]|}(E_K(\mathsf{inc}^i(\mathsf{Ctr0})))$
7. $C \leftarrow C[1] \,\|\, \cdots \,\|\, C[m-1] \,\|\, C[m]$
8. **return** C

Algorithm $\mathsf{GMAC}.\mathcal{T}[E_K](N, A, C)$

1. $L \leftarrow E_K(0^n)$
2. **if** $\nu = 96$ **then** $\mathsf{Ctr0} \leftarrow N \,\|\, 0^{31}1$
3. **else then** $\mathsf{Ctr0} \leftarrow \mathsf{GHASH}_L(\varepsilon, N)$
4. $\mathsf{Mask} \leftarrow \mathtt{msb}_\tau(E_K(\mathsf{Ctr0}))$
5. $T \leftarrow \mathsf{Mask} \oplus \mathtt{msb}_\tau(\mathsf{GHASH}_L(A, C))$
6. **return** T

Algorithm $\mathsf{GMAC}.\mathcal{V}[E_K](N, A, C, T)$

1. $\widehat{T} \leftarrow \mathsf{GMAC}.\mathcal{T}[E_K](N, A, C)$
2. **if** $T = \widehat{T}$ **return** $\top$
3. **else return** $\bot$

Algorithm $\mathsf{GHASH}_L(A, C)$

1. $X \leftarrow \mathsf{pad}_n(A) \,\|\, \mathsf{pad}_n(C) \,\|\, \langle |A| \rangle_{n/2} \,\|\, \langle |C| \rangle_{n/2}$
2. $(X[1], \ldots, X[x]) \xleftarrow{n} X,\ Y \leftarrow 0^n$
3. **for** $i = 1$ **to** x **do** $Y \leftarrow L \cdot (Y \oplus X[i])$
4. **return** Y

Fig. 1: Algorithms of $\mathsf{GCM}[E_K]$. Note that $\nu := |N|$ and $\tau := |T|$ are fixed in advance. The operation "$\cdot$" in line 3 of GHASH denotes a multiplication over $\mathrm{GF}(2^n)$.

$\sum_{i=0}^{\ell-1} x_i 2^i$. Let $\mathsf{inc}(X) = \mathtt{msb}_{n-32}(X) \,\|\, \langle \mathsf{int}(\mathtt{lsb}_{32}(X)) + 1 \mod 2^{32} \rangle_{32}$. For $i \geq 0$, $\mathsf{inc}^i(X)$ means that applying inc on X for i times. When $i = 0$, $\mathsf{inc}^i(X) = X$.

GCM is basically a composition of counter mode and a nonce-based MAC using a polynomial hashing over $\mathrm{GF}(2^n)$. Figure 1 shows the algorithms of GCM using E_K, denoted as $\mathsf{GCM}[E_K]$. In addition to the normal decryption, Fig. 1 shows the unverified decryption and verification routines together with GMAC MAC function inside GCM. They will be needed for our analysis. Here, $N, A, C \in \{0,1\}^*$ and $\tau := |T| \in [1..n]$ (NIST recommends $\tau \geq 64$). We also require $\nu := |N| \in [1..2^{n/2} - 1]$, $|A| \in [0..2^{n/2} - 1]$, and $|M| \in [0..n(2^{32} - 2)]$. The parameters ν and τ must be fixed in advance. The derivation of the first counter block input depends on whether $\nu = 96$ or not. Note that $|C| = |M|$ holds. Figure 2 depicts the encryption.

4.2 INT-RUP bounds of GCM

Let $\mathsf{GCM}[E] = (\mathsf{GCM}.\mathcal{E}[E], \mathsf{GCM}.u\mathcal{D}[E], \mathsf{GCM}.\mathcal{V}[E])$ employing a block cipher $E : \mathcal{K} \times \{0,1\}^n \rightarrow \{0,1\}^n$. Let $\mathcal{A}$ be an INT-RUP adversary who makes q_e queries to $\mathsf{GCM}.\mathcal{E}[E_K]$, q_d queries to $\mathsf{GCM}.u\mathcal{D}[E_K]$, and q_v queries to $\mathsf{GCM}.\mathcal{V}[E_K]$, with

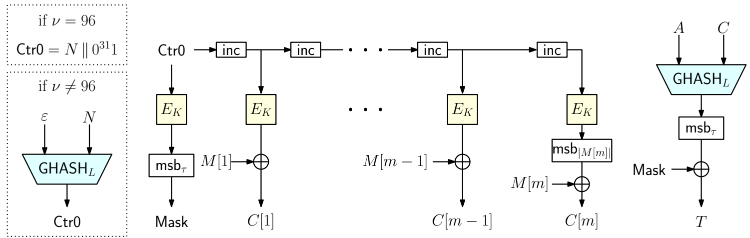

Fig. 2: Encryption of GCM.

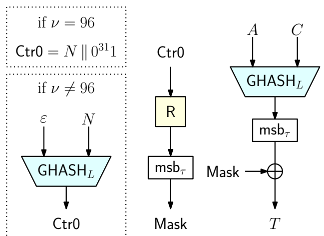

Fig. 3: $\mathsf{GMAC}.\mathcal{T}[\mathsf{R}]$.

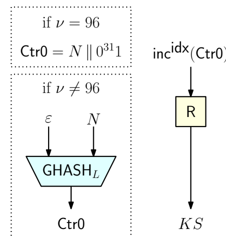

Fig. 4: $\mathcal{KS}_{\mathsf{G}}[\mathsf{R}]$, where $\mathsf{idx} \in [1..2^{32}-2]$.

time complexity t (where $K \xleftarrow{\$} \mathcal{K}$). After all the queries, $\mathcal{A}$ obtains a transcript written as

- $\{(N_i^e, A_i^e, M_i^e, C_i^e, T_i^e)\}_{i=1,\dots,q_e}$ from $\mathsf{GCM}.\mathcal{E}[E_K]$,
- $\{(N_i^d, A_i^d, M_i^d, C_i^d, T_i^d)\}_{i=1,\dots,q_d}$ from $\mathsf{GCM}.u\mathcal{D}[E_K]$,
- $\{(N_i^v, A_i^v, C_i^v, T_i^v, b_i)\}_{i=1,\dots,q_v}$ from $\mathsf{GCM}.\mathcal{V}[E_K]$, where $b_i \in \{\top, \bot\}$.

Let σ_e, σ_d, and σ_v be the total number of plaintext/ciphertext blocks queried to $\mathsf{GCM}.\mathcal{E}[E]$, $\mathsf{GCM}.u\mathcal{D}[E]$, and $\mathsf{GCM}.\mathcal{V}[E]$ oracles, *i.e.*, $\sigma_e = \sum_{i=1}^{q_e} |M_i^e|_n$, $\sigma_d = \sum_{i=1}^{q_d} |C_i^d|_n$, and $\sigma_v = \sum_{i=1}^{q_v} |C_i^v|_n$. Let $\sigma_{\text{all}} \coloneqq \sigma_e + \sigma_d + \sigma_v + q_e + q_v + 1$ to denote the maximum of the total number of E calls in the game, which we call the effective total queried blocks. Let $\ell_N \coloneqq \lceil \nu/n \rceil$ and ℓ_A be the maximum number of blocks of the input to GHASH in encryption and verification queries; thus, $|A_i^{\#}|_n + |C_i^{\#}|_n \le \ell_A$ for $\# \in \{e, v\}$, $i \in [1..q_{\#}]$.

The following theorem shows that GCM is INT-RUP secure up to the birthday bound irrespective of nonce length.

Theorem 1. *For the adversary $\mathcal{A}$ defined as above with time complexity t,*

$$\begin{aligned}
\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[E]}(\mathcal{A}) &\le \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} \\
&+ \frac{(q_e^2 + 2q_eq_v + 2q_e + 2q_v + 2\sigma_e + 2\sigma_d)(\ell_N + 1)}{2^n} \\
&+ \frac{64(2q_e + q_d + q_v - 1)(\sigma_e + \sigma_d + q_e + q_d)(\ell_N + 1)}{2^n} + \frac{2q_v(\ell_A + 2)}{2^\tau}
\end{aligned}$$

holds for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\text{all}})$. In particular, when $\nu = 96$, we have

$$\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[E]}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} + \frac{2q_v(\ell_A + 2)}{2^\tau}$$

for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\text{all}})$.

4.3 Proof overview

We show an overview of the proof for Theorem 1. The complete one is shown in App. B. First, we focus on the idealized $\mathsf{GCM}[\mathsf{P}]$ obtained by replacing E with a uniform random permutation P. This introduces the gap $\mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}})$ into the bound. We then apply the standard PRP/PRF switching lemma [20], thus, replacing all the invocations of P in the INT-RUP game of $\mathsf{GCM}[\mathsf{P}]$ with a uniform random function R, which leads to the term $\sigma_{\text{all}}^2/2^{n+1}$.

Second, to analyze $\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[\mathsf{R}]}(\mathcal{A}')$, we reduce the INT-RUP game of $\mathsf{GCM}[\mathsf{R}]$ to *a variant* of UF-CMA (Unforgeability under Chosen Message Attack) game of $\mathsf{GMAC}[\mathsf{R}]$. For this reduction, we first define the tagging oracle $\mathsf{GMAC}.\mathcal{T}[\mathsf{R}]$ and the verification oracle $\mathsf{GMAC}.\mathcal{V}[\mathsf{R}]$ as shown in Figs. 1, 3. In the standard UF-CMA game, the advantage is defined as the probability that the adversary querying $\mathsf{GMAC}.\mathcal{T}$ and $\mathsf{GMAC}.\mathcal{V}$ successfully obtains $\top$ from $\mathsf{GMAC}.\mathcal{V}$. Note that we assume the adversary cannot repeat nonces in the queries to $\mathsf{GMAC}.\mathcal{T}$. We then easily see that $\mathsf{GMAC}.\mathcal{T}$ (resp. $\mathsf{GMAC}.\mathcal{V}$) in the UF-CMA can simulate the tag generation part (resp. tag verification part) of $\mathsf{GCM}.\mathcal{E}$ (resp. $\mathsf{GCM}.\mathcal{V}$) in the INT-RUP game. To simulate the other part/oracle in the INT-RUP game of GCM, we additionally define a function; let $\mathcal{KS}_{\mathsf{G}}[\mathsf{R}] : \{0,1\}^\nu \times [1..2^{32}-2] \to \{0,1\}^n$ be a function that outputs a block of keystream $KS \in \{0,1\}^n$ for the tuple of $N \in \{0,1\}^\nu$ and block index $\mathsf{idx} \in [1..2^{32}-2]$ using R. Concretely, we define $\mathcal{KS}_{\mathsf{G}}[\mathsf{R}](N, \mathsf{idx}) \coloneqq \mathsf{R}(\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}))$, where

$$\mathsf{Ctr0} \coloneqq \begin{cases} N \,\|\, 0^{31}1 & \text{when} \quad \nu = 96 \\ \mathsf{GHASH}_L(\varepsilon, N) & \text{when} \quad \nu \neq 96, \end{cases}$$

and $L = \mathsf{R}(0^n)$. We also write it as $\mathcal{KS}_{\mathsf{G}}$. Figure 4 depicts $\mathcal{KS}_{\mathsf{G}}$. From the definition of $\mathcal{KS}_{\mathsf{G}}$, we see that $\mathcal{KS}_{\mathsf{G}}$ can simulate the keystream generation parts of $\mathsf{GCM}.\mathcal{E}/\mathsf{GCM}.\mathcal{V}$ and the whole of $\mathsf{GCM}.u\mathcal{D}$ in the INT-RUP. We then define a new UF-CMA-variant game, dubbed UF-CMA$^+$, involving $\mathsf{GMAC}.\mathcal{T}$, $\mathcal{KS}_{\mathsf{G}}$, and $\mathsf{GMAC}.\mathcal{V}$. Its definition is the UF-CMA game for $\mathsf{GMAC}[\mathsf{R}]$ with an additional oracle $\mathcal{KS}_{\mathsf{G}}$. For the detailed definition, see Def. 7 in App. B. From the above discussion, we can prove that the INT-RUP game of GCM can be simulated by the UF-CMA$^+$ game of GMAC and we obtain $\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[\mathsf{R}]}(\mathcal{A}') \leq \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B})$. See App. B.1 for the detailed proof.

Third, we analyze $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B})$ using an H-coefficient technique. In this part, we mainly show that a non-trivial collision probability between any of the inputs of R is sufficiently low. For this evaluation, we can utilize the evaluation in [62], which shows the upper bound of the collision probability in the standard authentication proof of GCM. Whereas our proof technique and the one used in [62] are different, and we additionally allow the adversary to query $\mathcal{KS}_{\mathsf{G}}$, we can prove that a similar bound as in the normal Auth of GCM [62].

5 Authenticity of CCM under RUP and Nonce Misuse, and Optimization of CCM

We show that CCM is NMR-INT-RUP secure. This also means CCM's INT-RUP and NMR-Auth security from the implications shown in Sect. 3. We also show that CCM maintains this robustness even if one omits a mask for the tag, which is derived by encrypting the nonce. We call this scheme $\mathsf{CCM2}$.

5.1 Specifications of CCM and CCM2

We describe CCM, partially adopting the notations of Gjiriti *et al.* [37]. As in the case of GCM, we use $\nu := |N|$ and $\tau := |T|$ and assume they are fixed in advance. CCM fixes $n = 128$ and $\nu \in \{56, 64, 72, 80, 88, 96, 104\}$ and $\tau \in \{32, 48, 64, 80, 96, 112, 128\}$. All input and output variables of CCM are assumed to be byte strings. A valid AD A and a valid plaintext M must satisfy $0 \leq |A| < (2^{64}) \cdot 8$ and $0 \leq |M| < (2^{120-\nu}) \cdot 8$. CCM composes a counter mode (which we write as CCTR) and $\mathsf{CBC\text{-}MAC}$ in a way similar to MAC-then-Encryption, toghther with a certain input encoding applied to $\mathsf{CBC\text{-}MAC}$.

Let $\mathsf{flags}(A)$ be a byte determined by AD A and ν and τ. As ν and τ are fixed, we do not write them as arguments. It is defined as $\mathsf{flags}(A) = (0 \,\|\, \mathsf{sign}(A = \varepsilon) \,\|\, \langle \tau/16 - 1\rangle_3 \,\|\, \langle 14 - \nu/8\rangle_3)$, where $\mathsf{sign}(\mathbf{S}) = 0$ if event $\mathbf{S}$ is true, $= 1$ otherwise. Let flags' be a byte $(0^5 \,\|\, \langle 14 - \nu/8\rangle_3)$. Due to the range of τ, $\mathsf{flags}' \neq \mathsf{flags}(A)$ holds for any $A \in \{0,1\}^*$. Let $\mathsf{ecN}(N, i) = (\mathsf{flags}' \,\|\, N \,\|\, \langle i\rangle_{120-\nu})$ denote the i-th counter block. $\mathsf{CBC\text{-}MAC}$ takes an encoded sequence, $B = (B[0], B[1], \ldots, B[\ell])$, generated from (N, A, M). In particular, let $\mathsf{ecB}_0(N, A, M) := (\mathsf{flags}(A) \,\|\, N \,\|\, \langle |M|_8\rangle_{120-\nu})$ which is used as $B[0]$. To determine the blocks after $B[0]$ in B, ecA encodes A as $\mathsf{ecA}(A) = \mathsf{pad}_n(\mathsf{lenA}(A) \,\|\, A)$. Here, $\mathsf{lenA}(A)$ denotes the length of A. Definitions of ecA and lenA are a bit complex and we omit the full details. See [33, 68]. Our proof will only use the following facts: (1) $|\mathsf{ecA}(A)|$ is a multiple of n bits for any valid A, (2) $\mathsf{ecA}(A) \neq \mathsf{ecA}(A')$ for any valid and distinct A and A', and (3) $|\mathsf{ecA}(A)|_n \leq |A|_n + 1$ for any A. Finally, the rest of B blocks are determined by the message encoding function $\mathsf{ecM}(M) := \mathsf{pad}_n(M)$. $\mathsf{CBC\text{-}MAC}$ takes a full-block sequence of $\mathsf{ec}(N, A, M) := \mathsf{ecB}_0(N, A, M) \,\|\, \mathsf{ecA}(A) \,\|\, \mathsf{ecM}(M)$. The τ-bit output of $\mathsf{CBC\text{-}MAC}$ taking $\mathsf{ec}(N, A, M)$ is masked by the most significant τ-bit of the first CCTR output with a counter block $\mathsf{ecN}(N, 0)$, which becomes a tag. The plaintext M is encrypted by CCTR taking counter blocks $\mathsf{ecN}(N, i)$ for $i = 1, \ldots, |M|_n$. See Fig. 5 for the pseudocode and Fig. 6 for the illustration. As with Sect. 4, Fig. 5 contains the unverified and the verification routines for our analysis.

The specification of $\mathsf{CCM2}$ is the same as that of CCM except that there is no mask for the tag. See Figs. 5, 6 for the definition.

Algorithm $\mathsf{CCM}.\mathcal{E}[E_K](N, A, M)$

$V \leftarrow \mathsf{CBC\text{-}MAC}.\mathcal{T}[E_K](N, A, M)$
$C \leftarrow \mathsf{CCTR}[E_K](N, M)$
$\boxed{U \leftarrow E_K(\mathsf{ecN}(N, 0))}$
$T \leftarrow V \ \boxed{\oplus \mathtt{msb}_\tau(U)}$
return (C, T)

Algorithm $\mathsf{CCM}.u\mathcal{D}[E_K](N, A, C, T)$

$M \leftarrow \mathsf{CCTR}[E_K](N, C)$
return M

Algorithm $\mathsf{CCM}.\mathcal{D}[E_K](N, A, C, T)$

$b = \mathsf{CCM}.\mathcal{V}[E_K](N, A, C, T)$
if $b = \bot$ **then return** $\bot$
else $M \leftarrow \mathsf{CCM}.u\mathcal{D}[E_K](N, A, C, T)$
return M

Algorithm $\mathsf{CCM}.\mathcal{V}[E_K](N, A, C, T)$

$M \leftarrow \mathsf{CCM}.u\mathcal{D}[E_K](N, A, C, T)$
$V \leftarrow \mathsf{CBC\text{-}MAC}.\mathcal{T}[E_K](N, A, M)$
$\boxed{U \leftarrow E_K(\mathsf{ecN}(N, 0))}$
$\widehat{T} \leftarrow V \ \boxed{\oplus \mathtt{msb}_\tau(U)}$
if $T = \widehat{T}$ **then return** $\top$
else return $\bot$

Algorithm $\mathsf{CBC\text{-}MAC}.\mathcal{T}[E_K](N, A, M)$
(assumes N, A and M are of valid length)

$V \leftarrow 0^n$
$B[0] \leftarrow \mathsf{ecB}_0(N, A, M)$
$(B[1], \ldots, B[\ell]) \xleftarrow{n} \mathsf{ecA}(A) \,\|\, \mathsf{ecM}(M)$
for $i = 0$ **to** ℓ
$\quad V \leftarrow E_K(V \oplus B[i])$
$V \leftarrow \mathtt{msb}_\tau(V)$
return V

Algorithm $\mathsf{CBC\text{-}MAC}.\mathcal{V}[E_K](N, A, M, V)$
(assumes N, A and M are of valid length and $|V| = \tau$)

$\widehat{V} \leftarrow \mathsf{CBC\text{-}MAC}.\mathcal{T}[E_K](N, A, M)$
if $V = \widehat{V}$ **return** $\top$
else return $\bot$

Algorithm $\mathsf{CCTR}[E_K](N, M)$

$(M[1], \ldots, M[m]) \xleftarrow{n} M$
for $i = 1$ **to** $m - 1$
$\quad C[i] \leftarrow E_K(\mathsf{ecN}(N, i)) \oplus M[i]$
$C[m] \leftarrow \mathtt{msb}_{|M[m]|}(E_K(\mathsf{ecN}(N, m))) \oplus M[m]$
$C \leftarrow C[1] \,\|\, \ldots \,\|\, C[m]$
return C

Fig. 5: Algorithms of CCM. $\nu := |N|$ and $\tau := |T|$ are fixed parameters. Algorithms of our optimization $\mathsf{CCM2}$ can be obtained by omitting gray boxes.

Definition 6 (Prefix-free encoding). *The encoding function* $\mathsf{ec}(\cdot, \cdot, \cdot)$ *is said to be prefix-free if, for any input* (N, A, M)*, there is no input* (N', A', M') *s.t.* $(N', A', M') \neq (N, A, M)$ *and* $\mathsf{ec}(N', A', M') = \mathtt{msb}_j(\mathsf{ec}(N, A, M))$*, where* $j = |\mathsf{ec}(N', A', M')|$ *and* $j \leq |\mathsf{ec}(N, A, M)|$*. We also say* $\{B = \mathsf{ec}(N, A, M)\}$ *is prefix-free for any input* (N, A, M) *if* $\mathsf{ec}(\cdot, \cdot, \cdot)$ *is prefix-free.*

An important observation is that ec in CCM is indeed prefix-free [47]. This was used by several studies for both attack and proof sides [36, 47]. It is known that $\mathsf{CBC\text{-}MAC}$ with a prefix-free input encoding realizes a PRF [65], however, CCTR and $\mathsf{CBC\text{-}MAC}$ in CCM share the key, and this shared-key structure does not allow to reuse this known result in a black-box way.

5.2 NMR-INT-RUP bound of CCM and CCM2

We focus on CCM since the bound of $\mathsf{CCM2}$ is almost the same as that of CCM. Let $\mathsf{CCM}[E_K] = (\mathsf{CCM}.\mathcal{E}[E_K], \mathsf{CCM}.u\mathcal{D}[E_K], \mathsf{CCM}.\mathcal{V}[E_K])$ for the underlying block cipher $E : \mathcal{K} \times \{0,1\}^n \to \{0,1\}^n$. We suppose the adversary $\mathcal{A}$ makes q_e queries to $\mathsf{CCM}.\mathcal{E}$, q_d queries to $\mathsf{CCM}.u\mathcal{D}$, and q_v queries to $\mathsf{CCM}.\mathcal{V}$. The adversary $\mathcal{A}$ obtains a transcript such that

– $\{(N_i^e, A_i^e, M_i^e, C_i^e, T_i^e)\}_{i=1,\ldots,q_e}$ from $\mathsf{CCM}.\mathcal{E}$,

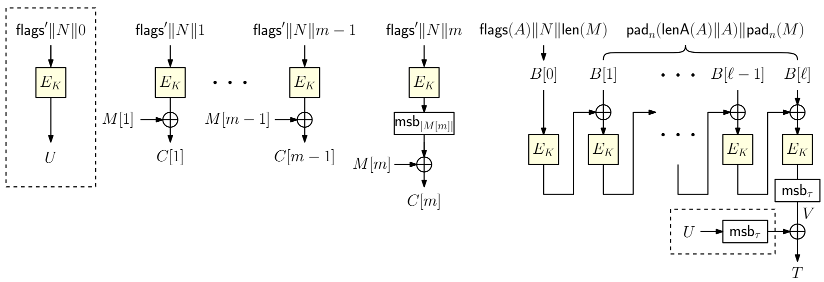

Fig. 6: $\mathsf{CCM}.\mathcal{E}[E_K]$. Our optimization ($\mathsf{CCM2}$) omits the boxes with dashed lines.

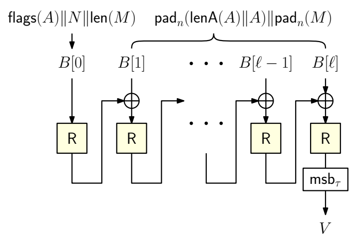

Fig. 7: $\mathsf{CBC\text{-}MAC}.\mathcal{T}[\mathsf{R}]$.

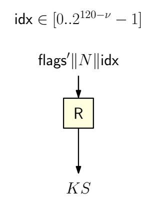

Fig. 8: $\mathcal{KS}_{\mathsf{C}}[\mathsf{R}]$.

– $\{(N_i^d, A_i^d, M_i^d, C_i^d, T_i^d)\}_{i=1,\ldots,q_d}$ from $\mathsf{CCM}.u\mathcal{D}$,
– $\{(N_i^v, A_i^v, C_i^v, T_i^v, b_i)\}_{i=1,\ldots,q_v}$ from $\mathsf{CCM}.\mathcal{V}$, where $b_i \in \{\top, \bot\}$.

Let σ_e, σ_d, and σ_v denote the total number of plaintext/ciphertext blocks queried in $\mathsf{CCM}.\mathcal{E}$, $\mathsf{CCM}u\mathcal{D}$, and $\mathsf{CCM}.\mathcal{V}$, respectively; thus, $\sigma_e = \sum_{i=1}^{q_e} |M_i^e|_n$, $\sigma_d = \sum_{i=1}^{q_d} |C_i^d|_n$, and $\sigma_v = \sum_{i=1}^{q_v} |C_i^v|_n$. Let σ_e' and σ_v' be the total number of input blocks to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ in $\mathsf{CCM}.\mathcal{E}$ and $\mathsf{CCM}.\mathcal{V}$, respectively; thus, $\sigma_e' = \sum_{i=1}^{q_e} |\mathsf{ec}(N_i^e, A_i^e, M_i^e)|_n$ and $\sigma_v' = \sum_{i=1}^{q_v} |\mathsf{ec}(N_i^v, A_i^v, M_i^v)|_n$, where $M_i^v = \mathsf{CCTR}(N_i^v, C_i^v)$. We define $\sigma_{\mathrm{all}} \coloneqq \sigma_e + \sigma_d + \sigma_v + \sigma_e' + \sigma_v' + q_e + q_v$ to denote the total number of block cipher calls in the game.

We present an NMR-INT-RUP bound of CCM. The bound shows up-to-birthday-bound security, hence quantitatively equivalent to the normal Auth bound [47].

Theorem 2. *Let $\mathcal{A}$ be the NMR-INT-RUP adversary against* $\mathsf{CCM}[E]$ *defined as above with time complexity t. We have*

$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM}[E]}(\mathcal{A}) \le \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\mathrm{all}}^2}{2^{n+1}} + \frac{(\sigma_e' + \sigma_v')(\sigma_e' + \sigma_v' + 2(q_e + q_v + \sigma_e + \sigma_d + \sigma_v))}{2^n} + \frac{4q_v}{2^\tau}$$

for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\mathrm{all}})$.

For the bound of $\mathsf{CCM2}$, the parameters can be determined in the same way as CCM, and we obtain the bound by replacing CCM and the third term of the bound in Theorem 2 with $\mathsf{CCM2}$ and $(\sigma'_e+\sigma'_v)(\sigma'_e+\sigma'_v+2(\sigma_e+\sigma_d+\sigma_v))/2^n$, respectively.

5.3 Proof overview

We show an overview of the NMR-INT-RUP proofs for both CCM and $\mathsf{CCM2}$. The complete ones are shown in App. C. As in the case of GCM, both proofs take two game hops: PRP/PRF switch and the security reduction to the UF-CMA$^+$ of the MAC part. By setting the MAC part as a pure $\mathsf{CBC\text{-}MAC}$ (not including the mask U), CCM and $\mathsf{CCM2}$'s NMR-INT-RUP bounds can be upper-bounded by the same UF-CMA$^+$ bound.

Let's consider $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}')$ and $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2[R]}}(\mathcal{A}')$, that appear after replacing E in $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM}[E]}(\mathcal{A})$ and $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2}[E]}(\mathcal{A})$ with P and then replace P with R. For the reduction to the UF-CMA$^+$ game, we define the tagging and the verification oracles of $\mathsf{CBC\text{-}MAC[R]}$, denoted by $\mathsf{CBC\text{-}MAC}.\mathcal{T}[\mathsf{R}]$ and $\mathsf{CBC\text{-}MAC}.\mathcal{V}[\mathsf{R}]$, as shown in Figs 5, 7. Additionally, we define a keystream oracle similar to $\mathcal{KS}_\mathsf{G}$. Let $\mathcal{KS}_\mathsf{C}[\mathsf{R}]$ be a function invoking R which takes (an encoded form of) a nonce $N\in\{0,1\}^\nu$ and an index $\mathsf{idx}\in[0..2^{120-\nu}-1]$ and outputs a block of keystream $KS\in\{0,1\}^n$. Specifically,

$$\mathcal{KS}_\mathsf{C}[\mathsf{R}](N,\mathsf{idx}) \coloneqq \mathsf{R}(\mathsf{ecN}(N,\mathsf{idx})) = \mathsf{R}(\mathsf{flags}' \parallel N \parallel \langle\mathsf{idx}\rangle_{120-\nu}).$$

We also write it as $\mathcal{KS}_\mathsf{C}$. Figure 8 depicts $\mathcal{KS}_\mathsf{C}$. We then define the UF-CMA$^+$ game involving $\mathsf{CBC\text{-}MAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{C}$, and $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ in the same manner as Sect. 4.3. Note that the adversary can repeat nonces in the queries to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$. From the definition, we observe that $\mathsf{CBC\text{-}MAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{C}$, and $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ in the UF-CMA$^+$ game can simulate $\mathsf{CCM}.\mathcal{E}$, $\mathsf{CCM}.u\mathcal{D}$, $\mathsf{CCM}.\mathcal{V}$ in the NMR-INT-RUP game, and the same holds for $\mathsf{CCM2}$. Thus, we obtain $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}') \le \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$ and $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2[R]}}(\mathcal{A}') \le \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$. See Lemma 6 and App. C.1 for the detailed statement and the proof.

Finally, we evaluate $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$. Intuitively, $\mathsf{CBC\text{-}MAC[R]}$ should be secure because the standard $\mathsf{CBC\text{-}MAC}$ is secure when its input is prefix-free, and in this case, $\{B=\mathsf{ec}(N,A,M)\}$ is prefix-free even in the nonce-misuse setting. However, the adversary in the UF-CMA$^+$ game also has access to $\mathcal{KS}_\mathsf{C}$. Therefore, we need a careful discussion, and for that, we use an H-coefficient technique. In the proof, as in the case of GCM, we mainly show that a non-trivial collision probability between any of the inputs of R is sufficiently low. The tricky point is to count *non-trivial* collisions. Trivial collisions can be invoked easily since $\mathsf{CBC\text{-}MAC}$ with a nonce-misuse setting can have two queries sharing the prefix. Careful discussion and case analysis of queries enable us to count non-trivial collisions, and we obtain a birthday bound, which is quantitatively the same as the normal Auth of CCM [47].

6 Privacy of CCM and CCM2 under Nonce Misuse

As Table 1 shows, CCM and $\mathsf{CCM2}$ do not fulfill NMR-Priv. However, CCM and $\mathsf{CCM2}$ still fulfill the weaker NML-Priv. The following theorem shows our bound.

Theorem 3. *Let q_r and q_m be the number of nonce-respecting and nonce-misusing queries of an adversary $\mathcal{A}$, and σ_{er} and σ_{em} be the total number of plaintext blocks queried to the nonce-respecting and nonce-misusing oracles. Let σ'_r and σ'_m be the total number of input blocks to underlying* CBC-MAC *in nonce-respecting and nonce-misusing queries. Let $\sigma_{\text{all}} \coloneqq \sigma'_r + \sigma'_m + \sigma_{er} + \sigma_{em} + q_r + q_m$ and t be the time complexity of $\mathcal{A}$. We obtain*

$$\mathbf{Adv}^{\texttt{NML-Priv}}_{\mathsf{CCM}[E]}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} + \frac{(\sigma'_r + \sigma'_m)(\sigma'_r + \sigma'_m + 2(\sigma_{er} + \sigma_{em} + q_r + q_m))}{2^n}$$

for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\text{all}})$. The bound of CCM2 *is obtained by replacing* $\mathsf{CCM}[E]$ *in the above equation with* $\mathsf{CCM2}[E]$ *and changing σ_{all} to $\sigma_{\text{all}} \coloneqq \sigma'_r + \sigma'_m + \sigma_{er} + \sigma_{em}$.*

The proof is similar to the NMR-INT-RUP proof of CCM/CCM2 described in Sect. 5.3 and App. C. Specifically, we define a new AE scheme, dubbed CCM+, that is the same as CCM, but $\mathsf{CCM+}.\mathcal{E}$ outputs (C, U, V) instead of (C, T). The NML-Priv bounds of CCM and CCM2 can be upper-bounded by the NML-Priv bound of CCM+. We can evaluate $\mathbf{Adv}^{\texttt{NML-Priv}}_{\mathsf{CCM+}}(\mathcal{A})$ similarly to $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$, which is analyzed in the last step of the NMR-INT-RUP proof of CCM/CCM2. See App. D for the full proof.

7 Nonce Misuse Resilience of OCB3

As shown by Table 1, OCB3 has no security with respect to RUP and NMR for both privacy and authenticity. What about NML security? ADL17 provided NML security analysis of OCB. Concretely, they showed NML-Priv and NML-Auth attacks against *the first version of* OCB, also known as OCB1 [73]. These attacks exploit the incomplete domain separation of OCB1 between the encryption of plaintext blocks and the encryption of checksum and recover $E_K(0^n)$ via repeating nonces. ADL17 stated, "We focus on OCB version 1 [65], however, our results extend to all versions of OCB" [16, Appendix A.1]. While there is no concrete interpretation of this claim, we do show that OCB3 is *not* completely broken regarding NML security.

Theorem 4. OCB3 *has NML-Priv security.*

Theorem 5. OCB3 *does not have NML-Auth security, however, if AD is fixed to any value throughout the game, it maintains NML-Auth security.*

The proofs of Theorems 4 and 5 are straightforward. To prove NML-Priv, we observe that OCB3 can be interpreted as a mode of TBC built on a block cipher, called ΘCB3 shown in Fig. 9 (App. E). It uses a TBC $\widetilde{E}$ based on XEX and XE modes by Rogaway [70] but with a more complex tweak space and mask derivation than the original (refer to [50] for the specification of $\widetilde{E}$). Thanks to the explicit domain separation via tweaks in $\Theta\mathsf{CB3}$, any output block in an encryption query with nonce N is of the form $\widetilde{E}_K^{(N,\mathtt{x})}(\alpha) \oplus \beta$ for some $\mathtt{x}$ (a tuple of size one or two) and some $\alpha, \beta \in \{0,1\}^n$ that do not involve computation of $\widetilde{E}_K^{(N,\mathtt{y})}$ for any possible $\mathtt{y}$. Suppose $\widetilde{E}_K$ is ideal (*i.e.*, a TURP $\widetilde{\mathsf{P}}$) and consider the real world in the NML-Priv game. The above observation implies that the responses from the first (nonce-respecting) oracle are completely random and independent of the responses from the second (nonce-misusing) oracles since the sets of used nonce values do not intersect. Therefore, $\Theta\mathsf{CB3}$ using $\widetilde{\mathsf{P}}$ is perfectly secure regarding NML-Priv. Thus, we obtain

$$\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{OCB3[P]}}(\mathcal{A}) = \mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\Theta\mathsf{CB3}[\widetilde{E}[\mathsf{P}]]}(\mathcal{A}) \leq \mathbf{Adv}^{\mathtt{TPRP}}_{\widetilde{E}[\mathsf{P}]}(\widehat{\mathcal{A}}) + \mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\Theta\mathsf{CB3}[\widetilde{\mathsf{P}}]}(\mathcal{A}) \leq \frac{6\sigma^2}{2^n}$$

for any NML-Priv adversary $\mathcal{A}$ with σ total queried blocks, for a TPRP adversary $\widehat{\mathcal{A}}$ with σ queries. Here, $\widetilde{E}[\mathsf{P}]$ is a TBC built on n-bit URP P, and the last inequality follows from [50, Lemma 3][6].

For NML-Auth, if no restriction on AD is imposed, Vaudenay and Vizár [78] showed an attack, which exploits the fact that AD processing of $\Theta\mathsf{CB3}/\mathsf{OCB3}$ does not involve nonce. In contrast, the attack generally does not work if A is fixed to any value. In $\Theta\mathsf{CB3}$, AD processing function, Hash, (see Fig. 9) uses distinct tweak values for the internal $\widetilde{\mathsf{P}}$ that are never used in the “core” encryption routine, G, because those used for Hash do not contain nonce while those for G always contain nonce. This fact implies that, if A is fixed to any value throughout the game, the encryption queries with repeating nonces do not contribute, hence can be ignored. Eventually, we can reuse the existing (nonce-respecting) Auth proof of $\Theta\mathsf{CB3}$ [50] including the bound. The final bound for $\mathsf{OCB3}$ will be obtained by adding the TBC's security bound [50, Lemma 3]. A proof is straightforward and hence omitted.

Some ways to achieve NML-Auth security exist. A simple option is to change Hash of $\Theta\mathsf{CB3}$ so that it takes nonce as a part of the tweak for each AD block process. This also allows the reuse of existing Auth proof even if AD is not fixed, making the modified $\mathsf{OCB3}$ NML secure for both privacy and authenticity. This comes at the cost of losing the property of efficient processing for static AD (*i.e.*, caching Hash(A)). Another option is to add Hash(A) to the checksum instead of the tag, *i.e.*, the tag is an encryption of the sum of checksum and Hash(A). This enables caching static AD, although the proof needs some revisions (future work).

[6] The TBC $\widetilde{E}$ reduces to (variants of) XEX and XE depending on the tweak used, and $\widetilde{E}$ is designed so that it is indistinguishable from TURP as long as the adversary performs CCA against XEX and performs only CPA against XE. This security property is needed for authenticity. We remark that $\mathsf{OCB2}$ [70] used a similar, but wrong, combination, which leads to an attack [44]. However, the case of $\mathsf{OCB3}$ is different and safe.

Table 2: Comparison of RUP-secure AEs based on GCM. F and G are PRFs; $F : \mathcal{K}' \times \{0,1\}^* \to \{0,1\}^\nu$ and $G : \mathcal{K}' \times \{0,1\}^\nu \to \{0,1\}^\nu$ for some key space $\mathcal{K}'$.

Scheme	Black-box	Additional cost	Online (Dec)	Output expansion$*$
PRF-to-IV + GCM	✓	$F_K(N \parallel A \parallel M)$	✓	ν
GCM-RUP [16]	✗	1 E_K + 1 GF(2^n) Mul †	✗	$n - \nu$
Nonce decoy + GCM	✓	$G_K(N)$	✓	ν

$*$ Additional output expansion in bits from the original GCM

† This can be none when the length of the last message block is in $[1..n - \tau]$.

8 Implications of our results

Proving robustness property is generally relevant as it gives defense in depth. Still, it is also natural to ask if there are real benefits. We show that this is indeed the case: our results enable 1) an efficient GCM-based RUP-secure AE and 2) an optimized version of CCM without losing security.

8.1 RUP-secure AE

An AE scheme equipped with PA1/2 and INT-RUP security properties is often called *RUP-secure* AE. It has been studied extensively [12,14,16,30]. Andreeva *et al.* [14] (ABL+14) shows *nonce decoy*, a generic method to turn an AE scheme $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ into a PA1-secure one if Π is PA1 secure under a random nonce setting. It takes an independently-keyed PRF, $G : \mathcal{K}' \times \{0,1\}^\nu \to \{0,1\}^\nu$, to encrypt the nonce N, and $G_{K'}(N)$ is given to $\Pi.\mathcal{E}$ as an input nonce. Nonce decoy preserves INT-RUP security of the baseline scheme if it has [14, Proposition 10]. ABL+14 also introduces another conversion scheme, called PRF-to-IV, from an AE scheme Π into a PA1 and INT-RUP secure one if Π is PA1 secure under a random nonce setting. While Π does not need to be INT-RUP secure, it also needs a variable-input-length PRF to take the whole input tuple of (N, A, M), imposing a significant computational overhead. ABL+14 shows that CTR is PA1 secure under a random nonce setting and that PA1 security of GCM and CCM can be reduced to that of CTR. Combining this result of ABL+14 and our INT-RUP results of GCM and CCM, nonce decoy with GCM or CCM implements RUP-secure AE *without modifying the algorithms of* GCM/CCM and *at a small computation overhead*. ADL17 introduces a RUP-secure variant of GCM, GCM-RUP, but this modifies the original algorithm, although the change is not extensive. Compared with GCM-RUP, nonce decoy works as a black-box conversion, enabling an easier adoption from the implementation point of view. Moreover, while the decryption of GCM-RUP cannot be online because of its MAC-then-Decrypt construction, the combination of nonce decoy and GCM preserves its online property. On the downside, as a nature of nonce decoy, the output must be expanded by ν bits from GCM. See the comparison in Table 2.

Table 3: Number of block cipher calls for encryption of a 128-bit block of message without AD. $(2+1)$ means the first two calls are in parallel followed by one call.

Scheme	GCM	CCM	OCB3	COFB	CCM2
Calls	3∗ (3)	4 (2+2)	4 (2+2)	3 (1+1+1)	3 (2+1)

* GCM needs additional GF(2^{128}) multiplications (2 when $\nu = 96$ and 4 when $\nu \in [1..128] \setminus \{96\}$)

8.2 Optimizing CCM

CCM2 reduces the number of block cipher calls by one for any input. The gain may seem minor at first sight, however, we do not know any example among the popular standards that allow such optimization after their standardization. In addition, this optimization may be relevant in some applications. Adomnicăi *et al.* [13] showed that CCM needs at least four block cipher calls (overhead) for any non-empty message. As far as known in the literature, three is the minimum number of overhead achievable by a general-purpose (*i.e.*, taking variable-length inputs) AE mode, namely COFB [17, 29]. See Table 3. Overhead is a critical measure to determine the performance for (very) short inputs. Consequently, CCM2 is yet another mode that achieves the minimum overhead of three calls, implying excellent performance for (very) short inputs, notably on microcontrollers [13].

9 Concluding Remarks

We have presented multiple new robustness properties for GCM, CCM, and OCB3. Our results provide a complete answer to the robustness of these standards and serve as a reference point for future research. Moreover, we showed that one block cipher call of CCM could be removed without harming robustness by presenting CCM2. Notably, CCM's strong robustness is somewhat surprising. This feature was probably not intended by the designers. We understand the design issues around CCM as pointed out by [68, 72]. Still, our results will give some positive views on this classical mode.

Studying the tightness of our bounds would be an interesting future topic. Extending the targets to the modern schemes, including Sponge AE schemes, would also be interesting. Finally, designing a strongly robust AE mode with minimal overhead (an early attempt is CLOC [45]), not limited to be based on CCM, is also worth investigating.

References

1. Ascon. https://ascon.iaik.tugraz.at/publications.html
2. Competition for authenticated encryption: Security, applicability, and robustness. https://competitions.cr.yp.to/caesar.html
3. Nist internal report 8454: Status report on the final round of the nist lightweight cryptography standardization process. https://nvlpubs.nist.gov/nistpubs/ir/2023/NIST.IR.8454.pdf

4. Nist lightweight cryptography. https://csrc.nist.gov/projects/lightweight-cryptography
5. Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC. NIST Special Publication 800-38D (2007), National Institute of Standards and Technology.
6. Recommendation for Block Cipher Modes of Operation: the CCM Mode for Authentication and Confidentiality. NIST Special Publication 800-38C (2007), National Institute of Standards and Technology.
7. The OCB Authenticated-Encryption Algorithm. IRTF RFC 7253 (2014)
8. Why is release of unverified plaintext so Bad? StackExchange (2018), https://crypto.stackexchange.com/questions/59973/why-is-release-of-unverified-plaintext-so-bad
9. How bad it is using the same IV twice with AES/GCM? StackExchange (2019), https://crypto.stackexchange.com/questions/26790/how-bad-it-is-using-the-same-iv-twice-with-aes-gcm
10. AES GCM : is it acceptable to return the wrong plaintext if the tag is incorrect? StackExchange (2021), https://crypto.stackexchange.com/questions/89365/aes-gcm-is-it-acceptable-to-return-the-wrong-plaintext-if-the-tag-is-incorrect
11. Abdelraheem, M.A., Beelen, P., Bogdanov, A., Tischhauser, E.: Twisted polynomials and forgery attacks on GCM. In: Oswald, E., Fischlin, M. (eds.) EUROCRYPT 2015, Part I. LNCS, vol. 9056, pp. 762–786. Springer, Heidelberg (Apr 2015). https://doi.org/10.1007/978-3-662-46800-5_29
12. Abed, F., Forler, C., List, E., Lucks, S., Wenzel, J.: RIV for robust authenticated encryption. In: Peyrin, T. (ed.) FSE 2016. LNCS, vol. 9783, pp. 23–42. Springer, Heidelberg (Mar 2016). https://doi.org/10.1007/978-3-662-52993-5_2
13. Adomnicai, A., Minematsu, K., Shikata, J.: Authenticated encryption for very short inputs. In: CT-RSA 2023. pp. 553–572. LNCS, Springer, Heidelberg (Feb 2023). https://doi.org/10.1007/978-3-031-30872-7_21
14. Andreeva, E., Bogdanov, A., Luykx, A., Mennink, B., Mouha, N., Yasuda, K.: How to securely release unverified plaintext in authenticated encryption. In: Sarkar, P., Iwata, T. (eds.) ASIACRYPT 2014, Part I. LNCS, vol. 8873, pp. 105–125. Springer, Heidelberg (Dec 2014). https://doi.org/10.1007/978-3-662-45611-8_6
15. Andreeva, E., Bogdanov, A., Luykx, A., Mennink, B., Tischhauser, E., Yasuda, K.: Parallelizable and authenticated online ciphers. In: Sako, K., Sarkar, P. (eds.) ASIACRYPT 2013, Part I. LNCS, vol. 8269, pp. 424–443. Springer, Heidelberg (Dec 2013). https://doi.org/10.1007/978-3-642-42033-7_22
16. Ashur, T., Dunkelman, O., Luykx, A.: Boosting authenticated encryption robustness with minimal modifications. In: Katz, J., Shacham, H. (eds.) CRYPTO 2017, Part III. LNCS, vol. 10403, pp. 3–33. Springer, Heidelberg (Aug 2017). https://doi.org/10.1007/978-3-319-63697-9_1
17. Banik, S., Chakraborti, A., Inoue, A., Iwata, T., Minematsu, K., Nandi, M., Peyrin, T., Sasaki, Y., Sim, S.M., Todo, Y.: GIFT-COFB. Submission to the NIST Lightweight Cryptography project (final round updates) (2022)
18. Baudrin, J., Canteaut, A., Perrin, L.: Practical cube attack against nonce-misused Ascon. IACR Trans. Symm. Cryptol. **2022**(4), 120–144 (2022). https://doi.org/10.46586/tosc.v2022.i4.120-144
19. Bellare, M., Namprempre, C.: Authenticated encryption: Relations among notions and analysis of the generic composition paradigm. In: Okamoto, T. (ed.) ASIACRYPT 2000. LNCS, vol. 1976, pp. 531–545. Springer, Heidelberg (Dec 2000). https://doi.org/10.1007/3-540-44448-3_41

20. Bellare, M., Rogaway, P.: The security of triple encryption and a framework for code-based game-playing proofs. In: Vaudenay, S. (ed.) EUROCRYPT 2006. LNCS, vol. 4004, pp. 409–426. Springer, Heidelberg (May / Jun 2006). https://doi.org/10.1007/11761679_25
21. Bellare, M., Tackmann, B.: The multi-user security of authenticated encryption: AES-GCM in TLS 1.3. In: Robshaw, M., Katz, J. (eds.) CRYPTO 2016, Part I. LNCS, vol. 9814, pp. 247–276. Springer, Heidelberg (Aug 2016). https://doi.org/10.1007/978-3-662-53018-4_10
22. Bellizia, D., Bronchain, O., Cassiers, G., Grosso, V., Guo, C., Momin, C., Pereira, O., Peters, T., Standaert, F.X.: Mode-level vs. implementation-level physical security in symmetric cryptography - A practical guide through the leakage-resistance jungle. In: Micciancio, D., Ristenpart, T. (eds.) CRYPTO 2020, Part I. LNCS, vol. 12170, pp. 369–400. Springer, Heidelberg (Aug 2020). https://doi.org/10.1007/978-3-030-56784-2_13
23. Berti, F., Guo, C., Pereira, O., Peters, T., Standaert, F.X.: TEDT: a leakage-resistant AEAD mode. IACR TCHES **2020**(1), 256–320 (2019). https://doi.org/10.13154/tches.v2020.i1.256-320, https://tches.iacr.org/index.php/TCHES/article/view/8400
24. Bhati, A.S., Andreeva, E., Vizár, D.: OAE-RUP: A strong online AEAD security notion and its application to SAEF. In: SCN (2). Lecture Notes in Computer Science, vol. 14974, pp. 117–139. Springer (2024)
25. Bhaumik, R., Nandi, M.: Improved security for OCB3. In: Takagi, T., Peyrin, T. (eds.) ASIACRYPT 2017, Part II. LNCS, vol. 10625, pp. 638–666. Springer, Heidelberg (Dec 2017). https://doi.org/10.1007/978-3-319-70697-9_22
26. Böck, H., Zauner, A., Devlin, S., Somorovsky, J., Jovanovic, P.: Nonce-Disrespecting Adversaries: Practical Forgery Attacks on GCM in TLS. In: WOOT. USENIX Association (2016)
27. Boldyreva, A., Degabriele, J.P., Paterson, K.G., Stam, M.: On symmetric encryption with distinguishable decryption failures. In: Moriai, S. (ed.) FSE 2013. LNCS, vol. 8424, pp. 367–390. Springer, Heidelberg (Mar 2014). https://doi.org/10.1007/978-3-662-43933-3_19
28. Bozhko, A.: Properties of AEAD Algorithms. Internet-Draft draft-irtf-cfrg-aead-properties-07, Internet Engineering Task Force (Jun 2024), https://datatracker.ietf.org/doc/draft-irtf-cfrg-aead-properties/07/, work in Progress
29. Chakraborti, A., Iwata, T., Minematsu, K., Nandi, M.: Blockcipher-based authenticated encryption: How small can we go? In: Fischer, W., Homma, N. (eds.) CHES 2017. LNCS, vol. 10529, pp. 277–298. Springer, Heidelberg (Sep 2017). https://doi.org/10.1007/978-3-319-66787-4_14
30. Chang, D., Datta, N., Dutta, A., Mennink, B., Nandi, M., Sanadhya, S., Sibleyras, F.: Release of unverified plaintext: Tight unified model and application to ANYDAE. IACR Trans. Symm. Cryptol. **2019**(4), 119–146 (2019). https://doi.org/10.13154/tosc.v2019.i4.119-146
31. Chen, S., Steinberger, J.P.: Tight security bounds for key-alternating ciphers. In: Nguyen, P.Q., Oswald, E. (eds.) EUROCRYPT 2014. LNCS, vol. 8441, pp. 327–350. Springer, Heidelberg (May 2014). https://doi.org/10.1007/978-3-642-55220-5_19
32. Dodis, Y., Grubbs, P., Ristenpart, T., Woodage, J.: Fast message franking: From invisible salamanders to encryptment. In: Shacham, H., Boldyreva, A. (eds.) CRYPTO 2018, Part I. LNCS, vol. 10991, pp. 155–186. Springer, Heidelberg (Aug 2018). https://doi.org/10.1007/978-3-319-96884-1_6

33. Dworkin, M.: Recommendation for Block Cipher Modes of Operation: the CCM Mode for Authentication and Confidentiality. NIST-SP 800-38C (2007)
34. Ferguson, N.: Authentication Weaknesses in GCM. Public Comments to NIST (2005), http://csrc.nist.gov/groups/ST/toolkit/BCM/comments.html
35. Fleischmann, E., Forler, C., Lucks, S.: McOE: A family of almost foolproof on-line authenticated encryption schemes. In: Canteaut, A. (ed.) FSE 2012. LNCS, vol. 7549, pp. 196–215. Springer, Heidelberg (Mar 2012). https://doi.org/10.1007/978-3-642-34047-5_12
36. Fouque, P.A., Martinet, G., Valette, F., Zimmer, S.: On the security of the CCM encryption mode and of a slight variant. In: Bellovin, S.M., Gennaro, R., Keromytis, A.D., Yung, M. (eds.) ACNS 08. LNCS, vol. 5037, pp. 411–428. Springer, Heidelberg (Jun 2008). https://doi.org/10.1007/978-3-540-68914-0_25
37. Gjiriti, E., Reyhanitabar, R., Vizár, D.: Power yoga: Variable-stretch security of CCM for energy-efficient lightweight IoT. IACR Trans. Symm. Cryptol. **2021**(2), 446–468 (2021). https://doi.org/10.46586/tosc.v2021.i2.446-468
38. Handschuh, H., Preneel, B.: Key-recovery attacks on universal hash function based MAC algorithms. In: Wagner, D. (ed.) CRYPTO 2008. LNCS, vol. 5157, pp. 144–161. Springer, Heidelberg (Aug 2008). https://doi.org/10.1007/978-3-540-85174-5_9
39. Hoang, V.T., Krovetz, T., Rogaway, P.: Robust authenticated-encryption AEZ and the problem that it solves. In: Oswald, E., Fischlin, M. (eds.) EUROCRYPT 2015, Part I. LNCS, vol. 9056, pp. 15–44. Springer, Heidelberg (Apr 2015). https://doi.org/10.1007/978-3-662-46800-5_2
40. Hoang, V.T., Reyhanitabar, R., Rogaway, P., Vizár, D.: Online authenticated-encryption and its nonce-reuse misuse-resistance. In: Gennaro, R., Robshaw, M.J.B. (eds.) CRYPTO 2015, Part I. LNCS, vol. 9215, pp. 493–517. Springer, Heidelberg (Aug 2015). https://doi.org/10.1007/978-3-662-47989-6_24
41. Hoang, V.T., Tessaro, S., Thiruvengadam, A.: The multi-user security of GCM, revisited: Tight bounds for nonce randomization. In: Lie, D., Mannan, M., Backes, M., Wang, X. (eds.) ACM CCS 2018. pp. 1429–1440. ACM Press (Oct 2018). https://doi.org/10.1145/3243734.3243816
42. Imamura, K., Minematsu, K., Iwata, T.: Integrity analysis of authenticated encryption based on stream ciphers. In: Chen, L., Han, J. (eds.) ProvSec 2016. LNCS, vol. 10005, pp. 257–276. Springer, Heidelberg (Nov 2016). https://doi.org/10.1007/978-3-319-47422-9_15
43. Imamura, K., Minematsu, K., Iwata, T.: Integrity analysis of authenticated encryption based on stream ciphers. Int. J. Inf. Sec. **17**(5), 493–511 (2018)
44. Inoue, A., Iwata, T., Minematsu, K., Poettering, B.: Cryptanalysis of OCB2: Attacks on authenticity and confidentiality. In: Boldyreva, A., Micciancio, D. (eds.) CRYPTO 2019, Part I. LNCS, vol. 11692, pp. 3–31. Springer, Heidelberg (Aug 2019). https://doi.org/10.1007/978-3-030-26948-7_1
45. Iwata, T., Minematsu, K., Guo, J., Morioka, S.: CLOC: Authenticated encryption for short input. In: Cid, C., Rechberger, C. (eds.) FSE 2014. LNCS, vol. 8540, pp. 149–167. Springer, Heidelberg (Mar 2015). https://doi.org/10.1007/978-3-662-46706-0_8
46. Iwata, T., Ohashi, K., Minematsu, K.: Breaking and repairing GCM security proofs. In: Safavi-Naini, R., Canetti, R. (eds.) CRYPTO 2012. LNCS, vol. 7417, pp. 31–49. Springer, Heidelberg (Aug 2012). https://doi.org/10.1007/978-3-642-32009-5_3

47. Jonsson, J.: On the security of CTR + CBC-MAC. In: Nyberg, K., Heys, H.M. (eds.) SAC 2002. LNCS, vol. 2595, pp. 76–93. Springer, Heidelberg (Aug 2003). https://doi.org/10.1007/3-540-36492-7_7
48. Joux, A.: Authentication Failures in NIST version of GCM. Public Comments to NIST (2006), http://csrc.nist.gov/groups/ST/toolkit/BCM/comments.html
49. KDDI Research, Inc.: Security Analysis of ChaCha20-Poly1305 AEAD. CRYPTREC-EX-2601-2016 (2017)
50. Krovetz, T., Rogaway, P.: The software performance of authenticated-encryption modes. In: Joux, A. (ed.) FSE 2011. LNCS, vol. 6733, pp. 306–327. Springer, Heidelberg (Feb 2011). https://doi.org/10.1007/978-3-642-21702-9_18
51. Lefevre, C., Mennink, B.: SoK: Security of the ascon modes. Cryptology ePrint Archive, Paper 2024/1969 (2024), https://eprint.iacr.org/2024/1969
52. Liénardy, J., Lafitte, F.: A weakness in OCB3 used with short nonces allowing for a break of authenticity and confidentiality. Inf. Process. Lett. **183**, 106404 (2024)
53. Liskov, M., Rivest, R.L., Wagner, D.: Tweakable block ciphers. In: Yung, M. (ed.) CRYPTO 2002. LNCS, vol. 2442, pp. 31–46. Springer, Heidelberg (Aug 2002). https://doi.org/10.1007/3-540-45708-9_3
54. Luykx, A., Mennink, B., Paterson, K.G.: Analyzing multi-key security degradation. In: Takagi, T., Peyrin, T. (eds.) ASIACRYPT 2017, Part II. LNCS, vol. 10625, pp. 575–605. Springer, Heidelberg (Dec 2017). https://doi.org/10.1007/978-3-319-70697-9_20
55. Luykx, A., Preneel, B.: Optimal forgeries against polynomial-based MACs and GCM. In: Nielsen, J.B., Rijmen, V. (eds.) EUROCRYPT 2018, Part I. LNCS, vol. 10820, pp. 445–467. Springer, Heidelberg (Apr / May 2018). https://doi.org/10.1007/978-3-319-78381-9_17
56. Mattsson, J., Westerlund, M.: Authentication key recovery on galois/counter mode (GCM). In: Pointcheval, D., Nitaj, A., Rachidi, T. (eds.) AFRICACRYPT 16. LNCS, vol. 9646, pp. 127–143. Springer, Heidelberg (Apr 2016). https://doi.org/10.1007/978-3-319-31517-1_7
57. McGrew, D.A., Viega, J.: The security and performance of the Galois/counter mode (GCM) of operation. In: Canteaut, A., Viswanathan, K. (eds.) INDOCRYPT 2004. LNCS, vol. 3348, pp. 343–355. Springer, Heidelberg (Dec 2004)
58. Menda, S., Len, J., Grubbs, P., Ristenpart, T.: Context discovery and commitment attacks - how to break CCM, EAX, SIV, and more. In: Hazay, C., Stam, M. (eds.) EUROCRYPT 2023, Part IV. LNCS, vol. 14007, pp. 379–407. Springer, Heidelberg (Apr 2023). https://doi.org/10.1007/978-3-031-30634-1_13
59. Mileva, A., Dimitrova, V., Velichkov, V.: Analysis of the Authenticated Cipher MORUS (v1). In: BalkanCryptSec. Lecture Notes in Computer Science, vol. 9540, pp. 45–59. Springer (2015)
60. Nandi, M.: Bernstein bound on WCS is tight - repairing luykx-preneel optimal forgeries. In: Shacham, H., Boldyreva, A. (eds.) CRYPTO 2018, Part II. LNCS, vol. 10992, pp. 213–238. Springer, Heidelberg (Aug 2018). https://doi.org/10.1007/978-3-319-96881-0_8
61. Nir, Y., Langley, A.: ChaCha20 and Poly1305 for IETF protocols. IRTF RFC 7539 (May 2015), https://tools.ietf.org/html/rfc7539
62. Niwa, Y., Ohashi, K., Minematsu, K., Iwata, T.: GCM security bounds reconsidered. In: Leander, G. (ed.) FSE 2015. LNCS, vol. 9054, pp. 385–407. Springer, Heidelberg (Mar 2015). https://doi.org/10.1007/978-3-662-48116-5_19
63. Patarin, J.: The "coefficients H" technique (invited talk). In: Avanzi, R.M., Keliher, L., Sica, F. (eds.) SAC 2008. LNCS, vol. 5381, pp. 328–345. Springer, Heidelberg (Aug 2009). https://doi.org/10.1007/978-3-642-04159-4_21

64. Pereira, O., Standaert, F.X., Vivek, S.: Leakage-resilient authentication and encryption from symmetric cryptographic primitives. In: Ray, I., Li, N., Kruegel, C. (eds.) ACM CCS 2015. pp. 96–108. ACM Press (Oct 2015). https://doi.org/10.1145/2810103.2813626
65. Petrank, E., Rackoff, C.: CBC MAC for real-time data sources. Journal of Cryptology **13**(3), 315–338 (Jun 2000). https://doi.org/10.1007/s001450010009
66. Poddebniak, D., Dresen, C., Müller, J., Ising, F., Schinzel, S., Friedberger, S., Somorovsky, J., Schwenk, J.: Efail: Breaking S/MIME and OpenPGP email encryption using exfiltration channels. In: Enck, W., Felt, A.P. (eds.) USENIX Security 2018. pp. 549–566. USENIX Association (Aug 2018)
67. Procter, G., Cid, C.: On weak keys and forgery attacks against polynomial-based MAC schemes. In: Moriai, S. (ed.) FSE 2013. LNCS, vol. 8424, pp. 287–304. Springer, Heidelberg (Mar 2014). https://doi.org/10.1007/978-3-662-43933-3_15
68. Rogaway, P., Wagner, D.: A critique of CCM. Cryptology ePrint Archive, Report 2003/070 (2003), https://eprint.iacr.org/2003/070
69. Rogaway, P.: Authenticated-encryption with associated-data. In: Atluri, V. (ed.) ACM CCS 2002. pp. 98–107. ACM Press (Nov 2002). https://doi.org/10.1145/586110.586125
70. Rogaway, P.: Efficient instantiations of tweakable blockciphers and refinements to modes OCB and PMAC. In: Lee, P.J. (ed.) ASIACRYPT 2004. LNCS, vol. 3329, pp. 16–31. Springer, Heidelberg (Dec 2004). https://doi.org/10.1007/978-3-540-30539-2_2
71. Rogaway, P.: Nonce-based symmetric encryption. In: Roy, B.K., Meier, W. (eds.) FSE 2004. LNCS, vol. 3017, pp. 348–359. Springer, Heidelberg (Feb 2004). https://doi.org/10.1007/978-3-540-25937-4_22
72. Rogaway, P.: Evaluation of Some Blockcipher Modes of Operation. Investigation Reports on Cryptographic Techniques in FY 2010 (2011), http://www.cryptrec.go.jp/english/
73. Rogaway, P., Bellare, M., Black, J., Krovetz, T.: OCB: A block-cipher mode of operation for efficient authenticated encryption. In: Reiter, M.K., Samarati, P. (eds.) ACM CCS 2001. pp. 196–205. ACM Press (Nov 2001). https://doi.org/10.1145/501983.502011
74. Rogaway, P., Shrimpton, T.: A provable-security treatment of the key-wrap problem. In: Vaudenay, S. (ed.) EUROCRYPT 2006. LNCS, vol. 4004, pp. 373–390. Springer, Heidelberg (May / Jun 2006). https://doi.org/10.1007/11761679_23
75. Sasaki, Y., Todo, Y., Aoki, K., Naito, Y., Sugawara, T., Murakami, Y., Matsui, M., Hirose, S.: Minalpher v1.1. Submission to the CAESAR competition (2015)
76. Vanhoef, M., Piessens, F.: Release the kraken: New KRACKs in the 802.11 standard. In: Lie, D., Mannan, M., Backes, M., Wang, X. (eds.) ACM CCS 2018. pp. 299–314. ACM Press (Oct 2018). https://doi.org/10.1145/3243734.3243807
77. Vaudenay, S.: Security flaws induced by CBC padding - applications to SSL, IPSEC, WTLS... In: Knudsen, L.R. (ed.) EUROCRYPT 2002. LNCS, vol. 2332, pp. 534–546. Springer, Heidelberg (Apr / May 2002). https://doi.org/10.1007/3-540-46035-7_35
78. Vaudenay, S., Vizár, D.: Can caesar beat galois? - Robustness of CAESAR candidates against nonce reusing and high data complexity attacks. In: Preneel, B., Vercauteren, F. (eds.) ACNS 18. LNCS, vol. 10892, pp. 476–494. Springer, Heidelberg (Jul 2018). https://doi.org/10.1007/978-3-319-93387-0_25
79. Zhang, X., Shen, Y., Wang, L.: Multi-user security of CCM authenticated encryption mode. In: CCS. pp. 4331–4345. ACM (2024)

Supplementary Material

A H-coefficient technique

We assume that an adversary $\mathcal{A}$ queries the two worlds, real and ideal, denoted by $\mathcal{O}_{\mathsf{re}}$ and $\mathcal{O}_{\mathsf{id}}$, and tries to distinguish them. The H-coefficient [31, 63] is a general technique to evaluate the distinguishing probability of $\mathcal{A}$. We define a *transcript* as a set of input/output values that $\mathcal{A}$ obtains during the interaction with the world. Let T_{re} (*resp.* T_{id}) denote the probability distribution of the transcript induced by the real world (*resp.* the ideal world). By extension, we also use the same notation to refer to a random variable distributed according to each distribution. We say that a transcript θ is *attainable* if $\Pr[\mathsf{T}_{\mathsf{id}} = \theta] > 0$ holds with respect to $\mathcal{A}$. Let Θ denote the set of attainable transcripts. The following is the fundamental lemma of the H-coefficient technique. See *e.g.* [31] for the proof.

Lemma 1. *Let $\Theta = \Theta_{\mathrm{good}} \sqcup \Theta_{\mathrm{bad}}$ be a partition of the set of attainable transcripts. Assume that there exists $\varepsilon_1 \geq 0$ such that for any $\theta \in \Theta_{\mathrm{good}}$, one has*

$$\frac{\Pr[\mathsf{T}_{\mathsf{re}} = \theta]}{\Pr[\mathsf{T}_{\mathsf{id}} = \theta]} \geq 1 - \varepsilon_1,$$

and that there exists $\varepsilon_2 \geq 0$ such that $\Pr[\mathsf{T}_{\mathsf{id}} \in \Theta_{\mathrm{bad}}] \leq \varepsilon_2$. Then $|\Pr[\mathcal{A}^{\mathcal{O}_{\mathsf{re}}} = 1] - \Pr[\mathcal{A}^{\mathcal{O}_{\mathsf{id}}} = 1]| \leq \varepsilon_1 + \varepsilon_2$.

B Proof of Theorem 1

Overview. We take three steps to prove INT-RUP bound for $\mathsf{GCM}[E]$. First, we apply PRP/PRF switching lemma [20] and replace P with a URF $\mathsf{R} : \{0,1\}^n \to \{0,1\}^n$. Second, we define a variant of UF-CMA (unforgeability under chosen message attack) security of $\mathsf{GMAC}[\mathsf{R}]$, dubbed UF-CMA$^+$, and show that INT-RUP security of $\mathsf{GCM}[\mathsf{R}]$ reduces to UF-CMA$^+$ security of $\mathsf{GMAC}[\mathsf{R}]$. Third, we prove UF-CMA$^+$ security of $\mathsf{GMAC}[\mathsf{R}]$ using the H-coefficient technique.

Step 1. An application of the standard PRP/PRF switching lemma shows

$$\mathbf{Adv}^{\mathtt{INT\text{-}RUP}}_{\mathsf{GCM}[E]}(\mathcal{A}) \leq \mathbf{Adv}^{\mathtt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\mathrm{all}}^2}{2^{n+1}} + \mathbf{Adv}^{\mathtt{INT\text{-}RUP}}_{\mathsf{GCM}[\mathsf{R}]}(\mathcal{A}') \qquad (1)$$

for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\mathrm{all}})$ and an INT-RUP adversary $\mathcal{A}'$ having the same computational cost as $\mathcal{A}$.

Step 2. We analyse $\mathbf{Adv}^{\mathtt{INT\text{-}RUP}}_{\mathsf{GCM}[\mathsf{R}]}(\mathcal{A}')$. Let $\mathsf{GMAC}[\mathsf{R}] = (\mathsf{GMAC}.\mathcal{T}[\mathsf{R}], \mathsf{GMAC}.\mathcal{V}[\mathsf{R}])$, where the algorithms of $\mathsf{GMAC}.\mathcal{T}$ and $\mathsf{GMAC}.\mathcal{V}$ are shown in Fig. 2. For simplicity, we write $\mathsf{GMAC}.\mathcal{T}$ and $\mathsf{GMAC}.\mathcal{V}$ to mean $\mathsf{GMAC}.\mathcal{T}[\mathsf{R}]$ and $\mathsf{GMAC}.\mathcal{V}[\mathsf{R}]$. Throughout this step, the same applies to other oracles, *e.g.*, $\mathsf{GCM}.u\mathcal{D}$ means

$\mathsf{GCM}.u\mathcal{D}[\mathsf{R}]$. We also define a function to simulate $\mathsf{GCM}.u\mathcal{D}$ and the encryption part of $\mathsf{GCM}.\mathcal{E}$ in the INT-RUP game. Let $\mathcal{KS}_\mathsf{G} : \{0,1\}^\nu \times [1..2^{32}-2] \to \{0,1\}^n$ be a function that outputs a block of keystream $KS \in \{0,1\}^n$ for the tuple of $N \in \{0,1\}^\nu$ and block index $\mathsf{idx} \in [1..2^{32}-2]$ using R. Concretely, we define $\mathcal{KS}_\mathsf{G}(N, \mathsf{idx}) \coloneqq \mathsf{R}(\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}))$, where

$$\mathsf{Ctr0} \coloneqq \begin{cases} N \,\|\, 0^{31}1 & \text{when } \ \nu = 96 \\ \mathsf{GHASH}_L(\varepsilon, N) & \text{when } \ \nu \neq 96, \end{cases}$$

and $L = \mathsf{R}(0^n)$. We define UF-CMA$^+$ game involving $\mathsf{GMAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{G}$, and $\mathsf{GMAC}.\mathcal{V}$. Its definition is UF-CMA game for $\mathsf{GMAC}[\mathsf{R}]$ with an additional oracle $\mathcal{KS}_\mathsf{G}$.

Definition 7 (UF-CMA$^+$ game). *Let $\mathcal{B}$ be an adversary querying to the three oracles, $\mathsf{GMAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{G}$, and $\mathsf{GMAC}.\mathcal{V}$. The adversary $\mathcal{B}$ does not forward queries from $\mathsf{GMAC}.\mathcal{T}$ to $\mathsf{GMAC}.\mathcal{V}$; $\mathcal{B}$ does not query (N, A, C, T) to $\mathsf{GMAC}.\mathcal{V}$ when $\mathcal{B}$ has already queried (N, A, C) to $\mathsf{GMAC}.\mathcal{T}$ and received T before. The UF-CMA$^+$ advantage of $\mathcal{B}$ is defined as*

$$\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B}) \coloneqq \Pr[\mathcal{B}^{\mathsf{GMAC}.\mathcal{T}, \mathcal{KS}_\mathsf{G}, \mathsf{GMAC}.\mathcal{V}} \text{ forges}],$$

*where the probability is defined over R and the random coins of $\mathcal{A}$. Here, "*forges*" means the event that $\mathsf{GMAC}.\mathcal{V}$ returns $\top$.*

In this section, we assume the adversary of UF-CMA$^+$ game $\mathcal{B}$ follows a nonce-respecting setting; thus, $\mathcal{B}$ does not repeat the same nonce in the queries to $\mathsf{GMAC}.\mathcal{T}$. We obtain the following lemma.

Lemma 2. *Let $\mathcal{A}'$ be an INT-RUP adversary against $\mathsf{GCM}[\mathsf{R}]$ using q_e encryption queries, q_d decryption queries, and q_v verification queries. Let σ_e and σ_d be as defined earlier. We have*

$$\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[\mathsf{R}]}(\mathcal{A}') \leq \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B})$$

for a UF-CMA$^+$ adversary $\mathcal{B}$ against $\mathsf{GMAC}[\mathsf{R}]$ that makes q_e queries to $\mathsf{GMAC}.\mathcal{T}$, $(\sigma_e + \sigma_d)$ queries to $\mathcal{KS}_\mathsf{G}$, and q_v queries to $\mathsf{GMAC}.\mathcal{V}$.

The proof of Lemma 2 is in App. B.1.

Step 3. We evaluate $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B})$. We define the following indistinguishability notion to apply the H-coefficient technique.

$$\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{GMAC}[\mathsf{R}]}(\mathcal{B}) \coloneqq |\Pr[\mathcal{B}^{\mathsf{GMAC}.\mathcal{T}, \mathcal{KS}_\mathsf{G}, \mathsf{GMAC}.\mathcal{V}} = 1] - \Pr[\mathcal{B}^{\$, \mathsf{R}_{ks}, \perp} = 1]|,$$

where \$ (resp. R_{ks}) is a random oracle (resp. URF) whose input/output interface and lengths are the same as $\mathsf{GMAC}.\mathcal{T}$ (resp. $\mathcal{KS}_\mathsf{G}$). Note that R_{ks} is independent of R used by $\mathsf{GMAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{G}$, and $\mathsf{GMAC}.\mathcal{V}$. The $\perp$ oracle has the same input interface as $\mathsf{GMAC}.\mathcal{V}$ and returns $\perp$ for any input. We obtain the following inequality.

$$\begin{aligned}\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC[R]}}(\mathcal{B}) &= |\Pr[\mathcal{B}^{\mathsf{GMAC}.\mathcal{T},\mathcal{KS}_{\mathsf{G}},\mathsf{GMAC}.\mathcal{V}} = 1] - \Pr[\mathcal{B}^{\mathsf{GMAC}.\mathcal{T},\mathcal{KS}_{\mathsf{G}},\perp} = 1] \\ &\quad + \Pr[\mathcal{B}^{\$,\mathsf{R}_{ks},\perp} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{R}_{ks},\perp} = 1]| \\ &\leq 2\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{GMAC[R]}}(\mathcal{B}). \end{aligned} \tag{2}$$

To bound the right-hand side of Eq. (2), we relax the game so that $\mathcal{B}$ obtains the GHASH key after all queries but before it determines an output bit. In the real world, $\mathcal{B}$ obtains a real key $L = \mathsf{R}(0^n)$, and in the ideal world, it obtains a dummy key $L \xleftarrow{\$} \{0,1\}^n$. Then, $\mathcal{B}$ can compute GHASH_L outputs for all queries. We define a transcript $\theta = (\theta_t, \theta_{ks}, \theta_v, L)$ of $\mathcal{B}$ such that

- $\theta_t = \{(N_i^t, A_i^t, C_i^t, T_i^t, \mathsf{Ctr0}_i^t, \mathsf{Mask}_i)\}_{i\in[1..q_t]}$,
- $\theta_{ks} = \{(N_i^{ks}, \mathsf{idx}_i, KS_i, \mathsf{Ctr0}_i^{ks})\}_{i\in[1..q_{ks}]}$,
- $\theta_v = \{(N_i^v, A_i^v, C_i^v, T_i^v, \mathsf{Ctr0}_i^v)\}_{i\in[1..q_v]}$,

where $\mathsf{Mask}_i = T_i^t \oplus \mathtt{msb}_\tau(\mathsf{GHASH}_L(A_i^t, C_i^t))$. Note that θ_v does not contain $b_i \in \{\top, \perp\}$ as any attainable transcript has $b_i = \perp$ for all $i \in [1..q_v]$ by definition. Let $\ell_N = \lceil \nu/n \rceil$ and ℓ_A be the maximum number of blocks of the input to GHASH in the game; thus, $|A_i^{\#}|_n + |C_i^{\#}|_n \leq \ell_A$ for $\# \in \{t, v\}$, $i \in [1..q_{\#}]$. For a transcript θ, we define the following bad events.

Bad1 Collision of $\mathsf{Ctr0}^t$ in θ_t: there exists distinct $i, j \in [1..q_t]$ *s.t.* $\mathsf{Ctr0}_i^t = \mathsf{Ctr0}_j^t$.

Bad2 Collision of $\mathsf{Ctr0}^t$ in θ_t and $\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}^{ks})$ in θ_{ks}: there exists $i \in [1..q_t]$, $j \in [1..q_{ks}]$ *s.t.* $\mathsf{Ctr0}_i^t = \mathsf{inc}^{\mathsf{idx}_j}(\mathsf{Ctr0}_j^{ks})$.

Bad3 Non-trivial collision of $\mathsf{Ctr0}^t$ in θ_t and $\mathsf{Ctr0}^v$ in θ_v: there exists $i \in [1..q_t]$, $j \in [1..q_v]$ *s.t.* $N_i^t \neq N_j^v$ and $\mathsf{Ctr0}_i^t = \mathsf{Ctr0}_j^v$.

Bad4 Non-trivial collision of $\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}^{ks})$ in θ_{ks}: there exists $i, j \in [1..q_{ks}]$ *s.t.* $(N_i^{ks}, \mathsf{idx}_i) \neq (N_j^{ks}, \mathsf{idx}_j)$ and $\mathsf{inc}^{\mathsf{idx}_i}(\mathsf{Ctr0}_i^{ks}) = \mathsf{inc}^{\mathsf{idx}_j}(\mathsf{Ctr0}_j^{ks})$.

Bad5 Collision of $\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}^{ks})$ in θ_{ks} and $\mathsf{Ctr0}^v$ in θ_v: there exists $i \in [1..q_v]$, $j \in [1..q_{ks}]$ *s.t.* $\mathsf{Ctr0}_i^v = \mathsf{inc}^{\mathsf{idx}_j}(\mathsf{Ctr0}_j^{ks})$.

Bad6 Collision of 0^n and any of $\mathsf{Ctr0}^t$, $\mathsf{inc}^{\mathsf{idx}}(\mathsf{Ctr0}^{ks})$, or $\mathsf{Ctr0}^v$ in θ: there exists $i \in [1..q_t]$, $j \in [1..q_{ks}]$, $k \in [1..q_v]$ *s.t.* $0^n = \mathsf{Ctr0}_i^t$ or $0^n = \mathsf{inc}^{\mathsf{idx}_j}(\mathsf{Ctr0}_j^{ks})$ or $0^n = \mathsf{Ctr0}_k^v$.

Bad7 Successful forgery: for $i \in [1..q_v]$, there exists $j \in [1..q_t]$ *s.t.* $N_i^v = N_j^t$ and $T_i^v = \mathsf{Mask}_j \oplus \mathtt{msb}_\tau(\mathsf{GHASH}_L(A_i^v, C_i^v))$.

We say (an attainable) θ is bad if any of the above bad events occur and define Θ_{bad} as the set of all bad transcripts. We define the good transcript set Θ_{good} following App. A. Let $\mathbf{Bad} \coloneqq \mathbf{Bad1} \cup \cdots \cup \mathbf{Bad7}$. We evaluate the upper bound of $\Pr[\mathsf{T}_{\mathsf{id}} \in \Theta_{\text{bad}}] =: \Pr[\mathbf{Bad}]$, where T_{id} is a random variable of the transcript in the ideal world.

Let us first focus on the general case of ν. We define q'_{ks} as the number of distinct queries of θ_{ks}. From the union bound for collisions among random elements and the fact that a polynomial of degree s has at most s solutions, we observe that **Bad1**, **Bad3**, **Bad6**, and **Bad7** are bounded as $\Pr[\mathbf{Bad1}] \leq \binom{q_t}{2}(\ell_N + 1)/2^n$, $\Pr[\mathbf{Bad3}] \leq q_t q_v(\ell_N + 1)/2^n$, $\Pr[\mathbf{Bad6}] \leq (q_t + q'_{ks} + q_v)(\ell_N + 1)/2^n$, and $\Pr[\mathbf{Bad7}] \leq q_v(\ell_A + 1)/2^\tau$. Note that the equations derived from these bad events are non-trivial equations of (random) L of degree at most $\ell_N + 1$ or $\ell_A + 1$ over $\mathrm{GF}(2^n)$. For the evaluation of **Bad2**, let $\mathsf{Coll}_L([a..b], N_1, N_2)$ be the event $\mathsf{GHASH}_L(\varepsilon, N_1) = \mathsf{inc}^r(\mathsf{GHASH}_L(\varepsilon, N_2))$ for some $r \in [a..b]$, where $0 \leq a \leq b \leq 2^{32} - 1$ and N_1 and N_2 are distinct nonces which are not 96 bits. The following lemma gives the upper bound of the event.

Lemma 3 (Lemmas 2 and 3 in [62]). *For $0 \leq m \leq 2^{32} - 1$ and two distinct nonces N_1 and N_2 which are not 96 bits, $\Pr[\mathsf{Coll}_L([0..m], N_1, N_2)] \leq 32(m + 1)(\ell_N + 1)/2^n$, where $|N_1|_n, |N_2|_n \leq \ell_N$ holds.*

To use Lemma 3, we define d as the number of distinct nonces in queries of θ_{ks}. For $i \in [1..d]$, let $\mathsf{idx}_i^{\mathsf{M}}$ be the maximum value of idx among queries for the i-th nonce N_i^{ks*}, using the arbitrary ordering of the nonces. Let $\rho = \sum_{i=1}^{d} \mathsf{idx}_i^{\mathsf{M}}$, and note that $q'_{ks} \leq \rho$. We then obtain the following evaluation.

$$\begin{aligned}\Pr[\mathbf{Bad2}] &\leq \sum_{i=1}^{q_t} \sum_{j=1}^{d} \Pr[\mathsf{Coll}_L([0..\mathsf{idx}_j^{\mathsf{M}}], N_i^t, N_j^{ks*})] \\ &\leq \sum_{i=1}^{q_t} \sum_{j=1}^{d} \frac{32(\mathsf{idx}_j^{\mathsf{M}} + 1)(\ell_N + 1)}{2^n} \leq \frac{32 q_t(\rho + d)(\ell_N + 1)}{2^n}.\end{aligned}$$

Note that $\Pr[\mathsf{Ctr0}_i^t = \mathsf{inc}^k(\mathsf{Ctr0}_j^{ks})] = 0$ when $N_i^t = N_j^{ks}$ since $k \geq 1$. We can evaluate $\Pr[\mathbf{Bad4}]$ and $\Pr[\mathbf{Bad5}]$ in the same manner as $\Pr[\mathbf{Bad2}]$.

$$\begin{aligned}\Pr[\mathbf{Bad5}] &\leq \frac{32 q_v(\rho + d)(\ell_N + 1)}{2^n}, \\ \Pr[\mathbf{Bad4}] &\leq \sum_{i=1}^{d} \sum_{\substack{j \neq i \\ j=1}}^{d} \frac{32(\mathsf{idx}_j^{\mathsf{M}} + 1)(\ell_N + 1)}{2^n} \leq \sum_{i=1}^{d} \frac{32(\rho - \mathsf{idx}_i^{\mathsf{M}} + d - 1)(\ell_N + 1)}{2^n} \\ &\leq \frac{32(d\rho - \rho + d^2 - d)(\ell_N + 1)}{2^n} \leq \frac{32(d-1)(\rho + d)(\ell_N + 1)}{2^n}.\end{aligned}$$

Summing up all the bad event probabilities, we obtain the following upper bound.

$$\begin{aligned}\Pr[\mathbf{Bad}] \leq &\frac{(0.5 q_t^2 + q_t q_v + q_t + q'_{ks} + q_v)(\ell_N + 1)}{2^n} \\ &+ \frac{32(q_t + q_v + d - 1)(\rho + d)(\ell_N + 1)}{2^n} + \frac{q_v(\ell_A + 1)}{2^\tau}. \qquad (3)\end{aligned}$$

Eq. (3) holds for any ν. On the other hand, when we fix $\nu = 96$, we obtain $\Pr[\mathbf{Bad1}] = \Pr[\mathbf{Bad2}] = \cdots = \Pr[\mathbf{Bad6}] = 0$, and

$$\Pr[\mathbf{Bad}] = \Pr[\mathbf{Bad7}] \leq \frac{q_v(\ell_A + 1)}{2^\tau}. \tag{4}$$

Next, we evaluate the lower bound of $\Pr[\mathsf{T}_{\mathsf{re}} = \theta]/\Pr[\mathsf{T}_{\mathsf{id}} = \theta]$ for $\theta \in \Theta_{\text{good}}$, where T_{re} is a random variable of the transcript in the real world. Let $\# \in \{\mathsf{re}, \mathsf{id}\}$. Let $\mathsf{T}^t_\#$, $\mathsf{T}^{ks}_\#$, $\mathsf{T}^v_\#$, $\mathsf{T}^L_\#$ denote the random variables of θ_t, θ_{ks}, θ_v, L in each world, respectively. For a good transcript $\theta \in \Theta_{\text{good}}$, we have

$$\begin{aligned}\Pr[\mathsf{T}_\# = \theta] = \Pr[\mathsf{T}^L_\# = L] \cdot \Pr[\mathsf{T}^{ks}_\# = \theta_{ks} \mid \mathsf{T}^L_\# = L] \\ \cdot \Pr[\mathsf{T}^t_\# = \theta_t \mid (\mathsf{T}^L_\#, \mathsf{T}^{ks}_\#) = (L, \theta_{ks})] \\ \cdot \Pr[\mathsf{T}^v_\# = \theta_v \mid (\mathsf{T}^t_\#, \mathsf{T}^L_\#, \mathsf{T}^{ks}_\#) = (\theta_t, L, \theta_{ks})].\end{aligned}$$

In the ideal world, we obtain

$$\Pr[\mathsf{T}_{\mathsf{id}} = \theta] = \left(\frac{1}{2^n}\right) \cdot \left(\frac{1}{2^n}\right)^{q'_{ks}} \cdot \left(\frac{1}{2^\tau}\right)^{q_t}. \tag{5}$$

Note that $\Pr[\mathsf{T}^v_{\mathsf{id}} = \theta_v \mid (\mathsf{T}^t_{\mathsf{id}}, \mathsf{T}^L_{\mathsf{id}}, \mathsf{T}^{ks}_{\mathsf{id}}) = (\theta_t, L, \theta_{ks})] = 1$ holds since the adversary always obtains $\perp$ in the ideal world. In the real world, we also obtain

$$\Pr[\mathsf{T}^L_{\mathsf{re}} = L] = \frac{1}{2^n}, \tag{6}$$

$$\Pr[\mathsf{T}^{ks}_{\mathsf{re}} = \theta_{ks} \mid \mathsf{T}^L_{\mathsf{re}} = L] = \left(\frac{1}{2^n}\right)^{q'_{ks}}, \tag{7}$$

$$\Pr[\mathsf{T}^t_{\mathsf{re}} = \theta_t \mid (\mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (L, \theta_{ks})] = \left(\frac{1}{2^\tau}\right)^{q_t}, \tag{8}$$

since θ is good. To be more precise, Eq. (7) holds since **Bad4** and **Bad6** do not happen, and Eq. (8) holds since **Bad1**, **Bad2**, and **Bad6** do not happen. All that remains is evaluating $\Pr[\mathsf{T}^v_{\mathsf{re}} = \theta_v \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})]$. For $i \in [1..q_v]$, let $\widehat{T}_i$ be a valid tag for the i-th verification query (N^v_i, A^v_i, C^v_i); thus, $\widehat{T}_i = \mathtt{msb}_\tau(\mathsf{R}(\mathsf{Ctr0}^v_i) \oplus \mathsf{GHASH}_L(A^v_i, C^v_i))$. We then obtain

$$\begin{aligned}&\Pr[\mathsf{T}^v_{\mathsf{re}} = \theta_v \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})] \\ &= \Pr[\{T^v_i \neq \widehat{T}_i\}_{i \in [1..q_v]} \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})] \\ &\geq 1 - \sum_{i=1}^{q_v} \Pr[T^v_i = \widehat{T}_i \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})] \geq 1 - \frac{q_v}{2^\tau}.\end{aligned} \tag{9}$$

Here, Eq. (9) holds because $\Pr[T^v_i = \widehat{T}_i \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})] \leq 1/2^\tau$ holds; when there exists $j \in [1..q_t]$ *s.t.* $N^v_i = N^t_j$, we obtain $\Pr[T^v_i = \widehat{T}_i \mid (\mathsf{T}^t_{\mathsf{re}}, \mathsf{T}^L_{\mathsf{re}}, \mathsf{T}^{ks}_{\mathsf{re}}) = (\theta_t, L, \theta_{ks})] = 0$ since **Bad7** does not happen. Otherwise, we obtain $\Pr[T_i^v = \widehat{T}_i \mid (\mathsf{T}_{\mathsf{re}}^t, \mathsf{T}_{\mathsf{re}}^L, \mathsf{T}_{\mathsf{re}}^{ks}) = (\theta_t, L, \theta_{ks})] \leq 1/2^\tau$ holds since **Bad3**, **Bad5**, and **Bad6** do not happen and $\mathsf{R}(\mathsf{Ctr0}_i^v)$ is uniformly random. From Eqs. (6), (7), (8), and (9), we obtain

$$\Pr[\mathsf{T}_{\mathsf{re}} = \theta] \geq \left(\frac{1}{2^n}\right) \cdot \left(\frac{1}{2^n}\right)^{q'_{ks}} \cdot \left(\frac{1}{2^\tau}\right)^{q_t} \cdot \left(1 - \frac{q_v}{2^\tau}\right). \tag{10}$$

Eqs. (5) and (10) show

$$\frac{\Pr[\mathsf{T}_{\mathsf{re}} = \theta]}{\Pr[\mathsf{T}_{\mathsf{id}} = \theta]} \geq 1 - \frac{q_v}{2^\tau}. \tag{11}$$

From Eqs. (3), (4), (11), and Lemma 1, we obtain the following lemma.

Lemma 4. *For any ν, we have*

$$\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{GMAC[R]}}(\mathcal{B}) \leq \frac{(0.5q_t^2 + q_t q_v + q_t + q'_{ks} + q_v)(\ell_N + 1)}{2^n} + \frac{32(q_t + q_v + d - 1)(\rho + d)(\ell_N + 1)}{2^n} + \frac{q_v(\ell_A + 2)}{2^\tau}.$$

In particular, when $\nu = 96$,

$$\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{GMAC[R]}}(\mathcal{B}) \leq \frac{q_v(\ell_A + 2)}{2^\tau}.$$

Applying Lemma 4 to Eq. (2) derives the upper bound of $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC[R]}}(\mathcal{B})$.

From the simulation of the INT-RUP game by the UF-CMA$^+$ adversary $\mathcal{B}$, we can assume $d \leq q_e + q_d$ and $\rho \leq \sigma_e + \sigma_d$ hold. Thus, we have

$$\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM[R]}}(\mathcal{A}) \leq \frac{(q_e^2 + 2q_e q_v + 2q_e + 2q_v + 2\sigma_e + 2\sigma_d)(\ell_N + 1)}{2^n} + \frac{64(2q_e + q_d + q_v - 1)(\sigma_e + \sigma_d + q_e + q_d)(\ell_N + 1)}{2^n} + \frac{2q_v(\ell_A + 2)}{2^\tau},$$

and when $\nu = 96$,

$$\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM[R]}}(\mathcal{A}) \leq \frac{2q_v(\ell_A + 2)}{2^\tau}.$$

Combining Eq. (1) and the above bounds concludes the proof.

Remark: alternative bound. In addition to Lemma 3, [62] shows an alternative bound of $\Pr[\mathsf{Coll}_L([0..m], N_1, N_2)]$. It has a larger constant than Lemma 3 but can be a smaller overall bound depending on the balance of the parameters. We discuss the use of another bound in App. B.2.

B.1 Proof of Lemma 2

We show the INT-RUP game of $\mathcal{A}$ can be simulated by $\mathcal{B}$ in the UF-CMA$^+$ game.

Encryption oracle: $\mathsf{GCM}.\mathcal{E}$ can be simulated using $\mathsf{GMAC}.\mathcal{T}$ and $\mathcal{KS}_\mathsf{G}$. Suppose that $\mathcal{A}$ queries (N^e, A^e, M^e) to $\mathsf{GCM}.\mathcal{E}$. To simulate the output of $\mathsf{GCM}.\mathcal{E}$, $\mathcal{B}$ queries $(N^e, 1), \ldots, (N^e, m^e)$ to $\mathcal{KS}_\mathsf{G}$, where $m^e = |M^e|_n$, and obtains their outputs $KS_1, \ldots, KS_{m^e}$. Then, $\mathcal{B}$ computes the ciphertext $C^e = M^e \oplus \mathtt{msb}_{|M^e|}(KS_1 \parallel \cdots \parallel KS_{m^e})$. The tag derivation can also be simulated by querying (N^e, A^e, C^e) to $\mathsf{GMAC}.\mathcal{T}$ and obtaining $T^e = \mathsf{GMAC}.\mathcal{T}(N^e, A^e, C^e)$. Thus, $\mathcal{B}$ can output (C^e, T^e) $(= \mathsf{GCM}.\mathcal{E}(N^e, A^e, M^e))$ with m^e queries to $\mathcal{KS}_\mathsf{G}$ and one query to $\mathsf{GMAC}.\mathcal{T}$. The total cost for q_e encryption queries to $\mathsf{GCM}.\mathcal{E}$ is σ_e queries to $\mathcal{KS}_\mathsf{G}$ and q_e queries to $\mathsf{GMAC}.\mathcal{T}$. Note that the above simulation maintains the nonce-respecting condition for UF-CMA$^+$ game as long as $\mathcal{A}$ is also nonce-respecting.

Unverified decryption oracle: $\mathsf{GCM}.u\mathcal{D}$ can be simulated using $\mathcal{KS}_\mathsf{G}$. Suppose that $\mathcal{A}$ queries (N^d, A^d, C^d, T^d) to $\mathsf{GCM}.u\mathcal{D}$. The adversary $\mathcal{B}$ queries $(N^d, 1)$, …, (N^d, m^d) to $\mathcal{KS}_\mathsf{G}$, where $m^d = |C^d|_n$, obtains their outputs $KS_1, \ldots, KS_{m^d}$, and outputs the unverified plaintext $M^d = C^d \oplus \mathtt{msb}_{|C^d|}(KS_1 \parallel \cdots \parallel KS_{m^d})$. The total cost for q_d encryption queries to $\mathsf{GCM}.u\mathcal{D}$ is at most σ_d queries to $\mathcal{KS}_\mathsf{G}$.

Verification oracle: $\mathsf{GCM}.\mathcal{V}$ can be simulated by simply forwarding its query to $\mathsf{GMAC}.\mathcal{V}$ since their algorithms are identical. This simulation does not invoke $\mathcal{B}$'s forwarding query from $\mathsf{GMAC}.\mathcal{T}$ to $\mathsf{GMAC}.\mathcal{V}$ since the forwarding queries from $\mathsf{GCM}.\mathcal{E}$ to $\mathsf{GCM}.\mathcal{V}$ are not allowed in INT-RUP game.

Therefore, the queries in INT-RUP game can be simulated by $\mathcal{B}$, and when $\mathcal{A}$ obtains $\top$ from $\mathsf{GCM}.\mathcal{V}$, $\mathcal{B}$ also obtains $\top$ from $\mathsf{GMAC}.\mathcal{V}$. This concludes the proof of Lemma 2.

B.2 Alternative bound for Theorem 1

In addition to Lemma 3, the authors of [62] show an alternative bound for the probability of $\mathsf{Coll}_L([0..m], N_1, N_2)$ useful for the case $\nu \neq 96$.

Lemma 5 (Lemmas 4 and 5 in [62]). *For $0 \leq m \leq 2^{32} - 1$ and two distinct nonces N_1 and N_2 which are not 96 bits, it holds that* $\Pr[\mathsf{Coll}_L([0..m], N_1, N_2)] \leq 2^{32}(\ell_N + 1)/2^n$*, where $|N_1|_n, |N_2|_n \leq \ell_N$.*

Lemma 5 could be used to derive another bound for $\mathbf{Adv}^{\mathtt{INT\text{-}RUP}}_{\mathsf{GCM[P]}}(\mathcal{A})$. While the above bound has a larger constant than that of Lemma 3, it is independent of m. Which lemma we should choose for bad event evaluations depends on the parameters. Specifically, when $(\sigma_e + \sigma_d)/(q_e + q_d) > 2^{27} - 1$, the following bound, based on Lemma 5, is smaller than that of Theorem 1. See [62, Sect. 6.5] for more details on the comparison.

Theorem 6. *For the same adversary $\mathcal{A}$ as in Theorem 1, we have*

$$\begin{aligned}
\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM}[E]}(\mathcal{A}) &\le \mathbf{Adv}^{\texttt{PRP}}_{E}(\mathcal{B}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} \\
&+ \frac{(q_e^2 + 2q_eq_v + 2q_e + 2q_v + 2\sigma_e + 2\sigma_d)(\ell_N + 1)}{2^n} \\
&+ \frac{2^{33}(2q_e + q_d + q_v - 1)(q_e + q_d)(\ell_N + 1)}{2^n} + \frac{2q_v(\ell_A + 2)}{2^\tau}
\end{aligned}$$

for a PRP adversary $\mathcal{B}$ with σ_{all} queries and time complexity $t + O(\sigma_{\text{all}})$.

Proof. The proof of Theorem 6 mostly follows the proof of Theorem 1 and [62]. From Lemma 5, we can reevaluate probabilities of **Bad2**, **Bad4**, and **Bad5** as follows.

$$\begin{aligned}
\Pr[\mathbf{Bad2}] &\le \sum_{i=1}^{q_t}\sum_{j=1}^{d} \Pr[\mathsf{Coll}_L([0..\mathsf{idx}^{\mathsf{M}}_j], N^t_i, N^{ks*}_j)] \le q_t d \frac{2^{32}(\ell_N + 1)}{2^n}, \\
\Pr[\mathbf{Bad4}] &\le \sum_{i=1}^{d}\sum_{\substack{j \ne i \\ j=1}}^{d} \Pr[\mathsf{Coll}_L([0..\mathsf{idx}^{\mathsf{M}}_j], N^{ks*}_i, N^{ks*}_j)] \le d(d-1) \frac{2^{32}(\ell_N + 1)}{2^n}, \\
\Pr[\mathbf{Bad5}] &\le \sum_{i=1}^{q_v}\sum_{j=1}^{d} \Pr[\mathsf{Coll}_L([0..\mathsf{idx}^{\mathsf{M}}_j], N^v_i, N^{ks*}_j)] \le q_v d \frac{2^{32}(\ell_N + 1)}{2^n}.
\end{aligned}$$

We then obtain the following evaluations in the same manner as the previous proof.

$$\begin{aligned}
\Pr[\mathbf{Bad}] &\le \frac{(0.5q_t^2 + q_tq_v + q_t + q'_{ks} + q_v)(\ell_N + 1)}{2^n} \\
&\quad + \frac{2^{32}d(q_t + q_v + d - 1)(\ell_N + 1)}{2^n} + \frac{q_v(\ell_A + 1)}{2^\tau}, \\
\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{GMAC[R]}}(\mathcal{B}) &\le \frac{(0.5q_t^2 + q_tq_v + q_t + q'_{ks} + q_v)(\ell_N + 1)}{2^n}, \\
&\quad + \frac{2^{32}d(q_t + q_v + d - 1)(\ell_N + 1)}{2^n} + \frac{q_v(\ell_A + 2)}{2^\tau} \\
\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{GMAC[R]}}(\mathcal{B}) &\le \frac{(q_t^2 + 2q_tq_v + 2q_t + 2q'_{ks} + 2q_v)(\ell_N + 1)}{2^n}, \\
&\quad + \frac{2^{33}d(q_t + q_v + d - 1)(\ell_N + 1)}{2^n} + \frac{2q_v(\ell_A + 2)}{2^\tau}, \\
\mathbf{Adv}^{\texttt{INT-RUP}}_{\mathsf{GCM[R]}}(\mathcal{A}) &\le \frac{(q_e^2 + 2q_eq_v + 2q_e + 2q_v + 2\sigma_e + 2\sigma_d)(\ell_N + 1)}{2^n}, \\
&\quad + \frac{2^{33}(q_e + q_d)(2q_e + q_d + q_v - 1)(\ell_N + 1)}{2^n} + \frac{2q_v(\ell_A + 2)}{2^\tau}.
\end{aligned}$$

Note that $q_t = q_e$, $q'_{ks} \le q_{ks} = \sigma_e + \sigma_d$, and $d \le q_e + q_d$.

C Proof of Theorem 2 and CCM2 bound

Overview. We evaluate $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM}[E]}(\mathcal{A})$ and $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2}[E]}(\mathcal{A})$ in the same manner as INT-RUP security proof of GCM. We first apply PRP/PRF switching lemma, show the reduction from NMR-INT-RUP game of CCM/CCM2 to UF-CMA$^+$ of CBC-MAC, and then prove $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$, where R is an n-bit URF. Note that the tag of CBC-MAC is the unmasked value, V, not the tag of CCM, T.

Step 1. We first focus on CCM and start with PRP/PRF switching applied to n-bit URP P and URF R. We have

$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM}[E]}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} + \mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}'), \tag{12}$$

for a PRP adversary $\widehat{\mathcal{A}}$ with σ_{all} queries and time complexity $t + O(\sigma_{\text{all}})$, and an NMR-INT-RUP adversary $\mathcal{A}'$ having the same computational cost as $\mathcal{A}$. We can evaluate $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2}[E]}(\mathcal{A})$ in the same manner as the above, and we obtain

$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2}[E]}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{PRP}}_{E}(\widehat{\mathcal{A}}) + \frac{\sigma_{\text{all}}^2}{2^{n+1}} + \mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2[R]}}(\mathcal{A}'). \tag{13}$$

Note that $\sigma_{\text{all}} \coloneqq \sigma_e + \sigma_d + \sigma_v + \sigma'_e + \sigma'_v$ in the case of CCM2.

Step 2. We evaluate $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}')$ and $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2[R]}}(\mathcal{A}')$. Let CBC-MAC[R] = (CBC-MAC.$\mathcal{T}$[R], CBC-MAC.$\mathcal{V}$[R]). See Figs. 5, 7 for the algorithms; note that CBC-MAC[R] is not a plain CBC-MAC but includes ec encoding function as a preprocessing of input tuple (N, A, M). The output of CBC-MAC.$\mathcal{T}$[R] is τ-bit string V. As in the proof in App. B, we hereafter may omit "[R]" for denoting the oracles/functions that invoke R, say we write CBC-MAC.$\mathcal{T}$ to mean CBC-MAC.$\mathcal{T}$[R]. We define a function to simulate CCM.$u\mathcal{D}$/CCM2.$u\mathcal{D}$, the encryption part of CCM.$\mathcal{E}$/CCM2.$\mathcal{E}$ (including the derivation of U), and the decryption part of CCM.$\mathcal{V}$/CCM2.$\mathcal{V}$ in the NMR-INT-RUP game. Let $\mathcal{KS}_{\mathsf{C}}$ be a function invoking R which takes (an encoded form of) a nonce $N \in \{0,1\}^{\nu}$ and an index $\mathsf{idx} \in [0..2^{120-\nu} - 1]$ and outputs a block of keystream $KS \in \{0,1\}^n$. Specifically,

$$\mathcal{KS}_{\mathsf{C}}(N, \mathsf{idx}) \coloneqq \mathsf{R}(\mathsf{ecN}(N, \mathsf{idx})) = \mathsf{R}(\mathsf{flags}' \,\|\, N \,\|\, \langle \mathsf{idx} \rangle_{120-\nu}).$$

We show that the NMR-INT-RUP game against CCM/CCM2 is simulatable with the UF-CMA$^+$ game against CBC-MAC defined in Def. 7. Note that, in this proof, we assume the adversary of the UF-CMA$^+$ game can repeat nonces in the queries to CBC-MAC.$\mathcal{T}$.

Lemma 6. *Let $\mathcal{A}$ be an NMR-INT-RUP adversary making q_e, q_d, and q_v queries to* CCM.$\mathcal{E}$, CCM.$u\mathcal{D}$, *and* CCM.$\mathcal{V}$ *respectively, and let σ_e, σ_d, σ_v, σ'_e, σ'_v be as defined earlier. We have*

$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$$

for a UF-CMA$^+$ adversary $\mathcal{B}$ against $\mathsf{CCM}[\mathsf{R}]$ *making* q_e, $q_e + q_v + \sigma_e + \sigma_d + \sigma_v$, *and* q_v *queries to* $\mathsf{CBC\text{-}MAC}.\mathcal{T}$, $\mathcal{KS}_\mathsf{C}$, $\mathsf{CBC\text{-}MAC}.\mathcal{V}$, *respectively. Here,* $\mathcal{B}$ *may repeat nonces for the first oracle.*

The proof of Lemma 6 is in App. C.1. The reduction on $\mathsf{CCM2}$ can be obtained by replacing CCM in Lemma 6 with $\mathsf{CCM2}$ and changing the number of queries to $\mathcal{KS}_\mathsf{C}$ to $\sigma_e + \sigma_d + \sigma_v$. The proof is almost identical to that of Lemma 6; thus, we omit it.

Step 3. To evaluate $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B})$, we define the following indistinguishability notion as in Sect. B.

$$\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B}) := |\Pr[\mathcal{B}^{\mathsf{CBC\text{-}MAC}.\mathcal{T},\mathcal{KS}_\mathsf{C},\mathsf{CBC\text{-}MAC}.\mathcal{V}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{R}_{ks},\perp} = 1]|,$$

where \$ (resp. R_{ks}) is a random oracle (resp. URF) whose input/output interface and lengths are the same as $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ (resp. $\mathcal{KS}_\mathsf{C}$). Note that R_{ks} is independent of the underlying random function R. Also, $\perp$ oracle has the same input interface as $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ and always returns $\perp$. As in Eq. (2), we obtain

$$\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B}) \leq 2\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B}). \tag{14}$$

We evaluate $\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B})$ using the H-coefficient technique. We suppose that $\mathcal{B}$ makes q_t tagging queries, q_{ks} keystream queries, and q_v verification queries. Let $(N_i^t, A_i^t, M_i^t, V_i^t)$, $(N_j^{ks}, \mathsf{idx}_j, KS_j)$, and $(N_k^v, A_k^v, M_k^v, V_k^v)$ be transcripts obtained from i-th tagging query for $i \in [1..q_t]$, j-th keystream query for $j \in [1..q_{ks}]$, and k-th verification query for $k \in [1..q_v]$, respectively. For $\# \in \{t, v\}$ and $i \in [1..q_\#]$, let $\mathsf{ec}(N_i^\#, A_i^\#, M_i^\#) = B_i^\# = (B_i^\#[0], \ldots, B_i^\#[\ell_i^\#])$, and $|B_i^\#|_n = \ell_i^\# + 1$. Let σ'_t and σ'_v be the total number of input blocks to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ and $\mathsf{CBC\text{-}MAC}.\mathcal{V}$; thus, for $\# \in \{t, v\}$, $\sigma'_\# := \sum_{i=1}^{q_\#}(\ell_i^\# + 1)$.

Simplification of the analysis. To simplify the analysis of $\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{CBC\text{-}MAC}[\mathsf{R}]}(\mathcal{B})$, we give $\mathcal{B}$ all the inputs to R in tagging and verification queries after all queries are made but before it determines an output bit. In the real world, this is straightforward. In the ideal world, we give dummy values, which are basically sampled from $\{0,1\}^n$ uniformly at random. One important point is that samplings must be *consistent* with queries. This part shows how to do that.

For $i \in [1..q_t]$ and $j \in [0..\ell_i^t]$, let $X_i^t[j]$ be $j+1$-th URF input of $\mathsf{CBC\text{-}MAC}$ in i-th tagging query, and for $i \in [1..q_v]$ and $j \in [0..\ell_i^v]$, let $X_i^v[j]$ be $j+1$-th URF input of $\mathsf{CBC\text{-}MAC}$ in i-th verification query. In the real world, for $\# \in \{t, v\}$, $i \in [1..q_\#]$, and $j \in [1..\ell_i^\#]$, we have

$$X_i^\#[0] = B_i^\#[0],$$
$$X_i^\#[j] = \mathsf{R}(X_i^\#[j-1]) \oplus B_i^\#[j].$$

Here, we list tagging and verification queries as $(N_1^t, A_1^t, M_1^t, V_1^t), \ldots, (N_{q_t}^t, A_{q_t}^t, M_{q_t}^t, V_{q_t}^t), (N_1^v, A_1^v, M_1^v, V_1^v), \ldots, (N_{q_v}^v, A_{q_v}^v, M_{q_v}^v, V_{q_v}^v)$. Thus, a *previous* query of i-th tagging query $(N_i^t, A_i^t, M_i^t, V_i^t)$ refers to j-th tagging query $(N_j^t, A_j^t, M_j^t, V_j^t)$ for

$j < i$. Also, a *previous* query of i-th verification query $(N_i^v, A_i^v, M_i^v, V_i^v)$ refers to j-th verification query $(N_j^v, A_j^v, M_j^v, V_j^v)$ or k-th tagging query $(N_k^t, A_k^t, M_k^t, V_k^t)$ for $j < i$ and $k \in [1..q_t]$.

For $\# \in \{t, v\}$, $i \in [1..q_\#]$, and $j \in [0..\ell_i^\#]$, we classify $X_i^\#[j]$ into the following three cases according to the value of $B_i^\#[0..j]$.

Case1 $X_i^\#[j]$ fulfills either of the following conditions:

(a) For $j = 0$, there are no previous queries *s.t.* $B_i^\#[0] = B_{i'}^{\#'}[0]$ for $\#' \in \{t, v\}$, $i' \in [1..q^\#]$.

(b) For $j \neq 0$, there are no previous queries *s.t.* $B_i^\#[0..j-1] = B_{i'}^{\#'}[0..j-1]$ for $\#' \in \{t, v\}$, $i' \in [1..q^\#]$.

Case2 There is a previous query *s.t.* $B_i^\#[0..j] = B_{i'}^{\#'}[0..j]$, and $X_{i'}^{\#'}[j]$ is classified as **Case1** for $\#' \in \{t, v\}$, $i' \in [1..q^\#]$.

Case3 For $j \neq 0$, $X_i^\#[j]$ fulfills both of the following conditions:

(a) $X_i^\#[j]$ is not classified as **Case2**.

(b) There is a previous query *s.t.* $|B_{i'}^{\#'}|_n \geq j+1$, $B_i^\#[0..j-1] = B_{i'}^{\#'}[0..j-1]$, $B_i^\#[j] \neq B_{i'}^{\#'}[j]$, and $X_{i'}^{\#'}[j]$ is classified as **Case1** for $\#' \in \{t, v\}$, $i' \in [1..q^\#]$.

Let $\mathcal{X}_\$$, $\mathcal{X}_=$, $\mathcal{X}_\oplus$ be a set of $X_i^\#[j]$ classified as **Case1**, **Case2**, and **Case3**, respectively. Technically, as each $X_i^\#[j]$ is a variable, each set is interpreted as a set of $(i, j, \#)$. Intuitively, $\mathcal{X}_\$$ denotes the set of (the indexes of) $X_i^\#[j]$ values that are to be sampled uniformly in the ideal world when $j \neq 0$, $\mathcal{X}_=$ denotes those are equal to a variable $X \in \mathcal{X}_\$$, and $\mathcal{X}_\oplus$ denotes those are determined by a sum of an element $X \in \mathcal{X}_\$$ and blocks in B. We obtain the following lemma.

Lemma 7. *Assume that* $\{B_i^\#\}_{\# \in \{t,v\}, i \in [1..q_\#]}$ *is prefix-free. Let* $\mathcal{X}_{all}$ *denote the set* $\{X_i^\#[j]\}_{\# \in \{t,v\}, i \in [1..q_\#], j \in [0..\ell_i^\#]}$. *Then the set of three sets,* $\mathcal{X}_\$$, $\mathcal{X}_=$, *and* $\mathcal{X}_\oplus$, *is a partition of* $\mathcal{X}_{all}$. *That is,* $\mathcal{X}_\$ \cap \mathcal{X}_= = \mathcal{X}_= \cap \mathcal{X}_\oplus = \mathcal{X}_\$ \cap \mathcal{X}_\oplus = \emptyset$ *holds and every* $X_i^\#[j]$ *is in any one of* $\mathcal{X}_\$$, $\mathcal{X}_=$, $\mathcal{X}_\oplus$.

The proof is a direct consequence of prefix-free encoding, hence we omit it. We note that, in order for Lemma 7 to hold, the prefix-freeness is necessary. For example, suppose that B_1^t is a prefix of B_2^t; say, suppose that $|B_2^t|_n = 3$ and there is a previous query *s.t.* $|B_1^t|_n = 2$ and $B_2^t[0..1] = B_1^t[0..1] = B_1^t$. In this case, $X_2^t[2]$ cannot be classified as any of the above three cases.

In this paper, we say $X_i^\#[j]$ is *a representative node*, when $X_i^\#[j] \in \mathcal{X}_\$ \sqcup \mathcal{X}_\oplus$. In the ideal world, when $j = 0$, the adversary can compute $X_i^\#[0]$ by itself. For $j \in [1..\ell_i^\#]$, the dummy value of $X_i^\#[j]$ is determined as follows.

1. When $X_i^\#[j] \in \mathcal{X}_\$$, $X_i^\#[j]$ is chosen from $\{0,1\}^n$ uniformly at random.
2. When $X_i^\#[j] \in \mathcal{X}_=$, $X_i^\#[j]$ is determined as $X_i^\#[j] = X_{i'}^{\#'}[j]$ using $X_{i'}^{\#'}[j]$ in a previous query, where $B_i^\#[0..j] = B_{i'}^{\#'}[0..j]$, and $X_{i'}^{\#'}[j] \in \mathcal{X}_\$$ for $\#' \in \{t, v\}$, $i' \in [1..q^\#]$.

3. When $X_i^{\#}[j] \in \mathcal{X}_\oplus$, $X_i^{\#}[j]$ is determined as $X_i^{\#}[j] = X_{i'}^{\#'}[j] \oplus B_{i'}^{\#'}[j] \oplus B_i^{\#}[j]$ using $X_{i'}^{\#'}[j]$ in a previous query, where $|B_{i'}^{\#'}|_n \geq j+1$, $B_i^{\#}[0..j-1] = B_{i'}^{\#'}[0..j-1]$, $B_i^{\#}[j] \neq B_{i'}^{\#'}[j]$, and $X_{i'}^{\#'}[j] \in \mathcal{X}_\$$ for $\#' \in \{t, v\}$, $i' \in [1..q^{\#}]$.

Definitions of a transcript and bad events. Including all the determined $X_i^{\#}[j]$, we define a transcript $\theta = (\theta_t, \theta_{ks}, \theta_v, \theta_x)$ of the adversary $\mathcal{B}$ such that

- $\theta_t = \{(N_i^t, A_i^t, M_i^t, V_i^t)\}_{i \in [1..q_t]}$,
- $\theta_{ks} = \{(N_i^{ks}, \mathsf{idx}_i, KS_i)\}_{i \in [1..q_{ks}]}$,
- $\theta_v = \{(N_i^v, A_i^v, M_i^v, V_i^v)\}_{i \in [1..q_v]}$,
- $\theta_x = \{X_i^{\#}[j]\}_{\# \in \{t,v\}, i \in [1..q_{\#}], j \in [0..\ell_i^{\#}]}$.

Note that we exclude $b_i \in \{\top, \bot\}$ in the verification queries from the above definition because an attainable transcript always has $b_i = \bot$.

A transcript θ is bad if any of the following bad events occur, and let Θ_{bad} be the set of bad transcripts. Let Θ_{good} denote the good transcript set following Appendix A.

Bad1 Collision of $X[\cdot]$ corresponding to representative nodes in θ_x: there exists $\#_1, \#_2 \in \{t, v\}$, $i_1 \in [1..q_{\#_1}]$, $i_2 \in [1..q_{\#_2}]$, $j_1 \in [0..\ell_{i_1}^{\#_1}]$, $j_2 \in [0..\ell_{i_2}^{\#_2}]$ *s.t.* $X_{i_1}^{\#_1}[j_1], X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\$ \sqcup \mathcal{X}_\oplus$ and $X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]$.

Bad2 Collision of $X[\cdot]$ corresponding to a representative node in θ_x and $\mathsf{ecN}(N^{ks}, \mathsf{idx})$ for some $(N^{ks}, \mathsf{idx}, KS) \in \theta_{ks}$: there exists $\# \in \{t, v\}$, $i \in [1..q_{\#}]$, $j \in [0..\ell_i^{\#}]$, $k \in [1..q_{ks}]$ *s.t.* $X_i^{\#}[j] \in \mathcal{X}_\$ \sqcup \mathcal{X}_\oplus$ and $X_i^{\#}[j] = \mathsf{ecN}(N_k^{ks}, \mathsf{idx}_k)$.

Bad3 Successful forgery: there exists $i \in [1..q_v]$ *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_j^t, A_j^t, M_j^t)$ and $V_i^v = V_j^t$ for $j \in [1..q_t]$.

Let **Bad** denote $\mathbf{Bad1} \cup \mathbf{Bad2} \cup \mathbf{Bad3}$. We evaluate $\Pr[\mathsf{T}_{\mathsf{id}} \in \Theta_{\mathrm{bad}}] =: \Pr[\mathbf{Bad}]$.

Bad1 evaluation. We fix $\#_1, \#_2 \in \{t, v\}$, $i_1 \in [1..q_{\#_1}]$, $i_2 \in [1..q_{\#_2}]$, $j_1 \in [0..\ell_{i_1}^{\#_1}]$, $j_2 \in [0..\ell_{i_2}^{\#_2}]$, $(\#_1, i_1, j_1) \neq (\#_2, i_2, j_2)$, *s.t.* both $X_{i_1}^{\#_1}[j_1]$ and $X_{i_2}^{\#_2}[j_2]$ are representative nodes.

First, we consider the case when $j_1 = 0$ or $j_2 = 0$. Without loss of generality, we can assume $j_1 = 0$. When $j_2 = 0$, we obtain $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = 0$ since they are representative nodes. When $j_2 \neq 0$ and $X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\$$, we obtain $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = 1/2^n$ since $X_{i_2}^{\#_2}[j_2]$ is randomly chosen. When $j_2 \neq 0$ and $X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\oplus$, we have $X_{i_2}^{\#_2}[j_2] = X_k^{\#'}[j_2] \oplus B_k^{\#'}[j_2] \oplus B_{i_2}^{\#_2}[j_2]$ for $\#' \in \{t, v\}$, $k \in [q_{\#'}]$, where $X_k^{\#'}[j_2] \in \mathcal{X}_\$$ from the definition of $\mathcal{X}_\oplus$. Thus, we obtain $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = \Pr[X_{i_1}^{\#_1}[j_1] = X_k^{\#'}[j_2] \oplus B_k^{\#'}[j_2] \oplus B_{i_2}^{\#_2}[j_2]] = 1/2^n$ since $X_k^{\#'}[j_2]$ is randomly chosen.

Second, we assume $j_1 \neq 0$ and $j_2 \neq 0$. We then consider the case when $X_{i_1}^{\#_1}[j_1] \in \mathcal{X}_\oplus$ or $X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\oplus$. We can assume $X_{i_1}^{\#_1}[j_1] \in \mathcal{X}_\oplus$ *w.l.o.g.* and let

$X_{i_1}^{\#_1}[j_1] = X_k^{\#'}[j_1] \oplus B_k^{\#'}[j_1] \oplus B_{i_1}^{\#_1}[j_1]$ for $\#' \in \{t, v\}$, $k \in [q_{\#'}]$, where $X_k^{\#'}[j_1] \in \mathcal{X}_\$$. Since $X_k^{\#'}[j_1]$ is randomly chosen, we obtain $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = \Pr[X_k^{\#'}[j_1] \oplus B_k^{\#'}[j_1] \oplus B_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] \leq 1/2^n$. Note that $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = 0$ holds when $\#' = \#_2$, $k = i_2$, $j_1 = j_2$ since $B_{i_2}^{\#_2}[j_1] \oplus B_{i_1}^{\#_1}[j_1] \neq 0$ holds from the definition of $\mathcal{X}_\oplus$.

Third, we assume $j_1 \neq 0$ and $j_2 \neq 0$. When both $X_{i_1}^{\#_1}[j_1]$, $X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\$$, $\Pr[X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]] = 1/2^n$ holds since they are randomly chosen.

Summing up all the cases of $(\#_1, i_1, j_1)$, $(\#_2, i_2, j_2)$, we obtain $\Pr[\mathbf{Bad1}] \leq (\sigma'_t + \sigma'_v)^2/2^{n+1}$.

Other bad events. We can evaluate **Bad2** in the same way as **Bad1**; $\Pr[\mathbf{Bad2}] \leq (\sigma'_t + \sigma'_v)q_{ks}/2^n$. Note that $\Pr[X_i^{\#}[0] = \mathsf{ecN}(N_k^{ks}, \mathsf{idx}_k)] = 0$ always holds due to the domain separation in the definition; *i.e.*, $\mathsf{flags}(A_i^{\#}) \neq \mathsf{flags}'$. Regarding **Bad3**, we obtain $\Pr[\mathbf{Bad3}] \leq q_v/2^\tau$ since V_j^t is randomly chosen from $\{0,1\}^\tau$.

Summing up all the bad event probabilities, we obtain

$$\Pr[\mathbf{Bad}] \leq \frac{(\sigma'_t + \sigma'_v)(0.5\sigma'_t + 0.5\sigma'_v + q_{ks})}{2^n} + \frac{q_v}{2^\tau}. \tag{15}$$

Good transcript ratio. We derive a lower bound of $\Pr[\mathsf{T}_{\mathsf{re}} = \theta]/\Pr[\mathsf{T}_{\mathsf{id}} = \theta]$ for $\theta \in \Theta_{\text{good}}$. Let $\# \in \{\mathsf{re}, \mathsf{id}\}$. Let $\mathsf{T}_\#^t$, $\mathsf{T}_\#^{ks}$, $\mathsf{T}_\#^v$, $\mathsf{T}_\#^x$ denote the random variables of θ_t, θ_{ks}, θ_v, θ_x in each world, respectively. For a good transcript $\theta \in \Theta_{\text{good}}$,

$$\begin{aligned}\Pr[\mathsf{T}_\# = \theta] = \Pr[\mathsf{T}_\#^{ks} = \theta_{ks}] &\cdot \Pr[\mathsf{T}_\#^x = \theta_x \mid \mathsf{T}_\#^{ks} = \theta_{ks}] \\ &\cdot \Pr[\mathsf{T}_\#^t = \theta_t \mid (\mathsf{T}_\#^{ks}, \mathsf{T}_\#^x) = (\theta_{ks}, \theta_x)] \\ &\cdot \Pr[\mathsf{T}_\#^v = \theta_v \mid (\mathsf{T}_\#^t, \mathsf{T}_\#^{ks}, \mathsf{T}_\#^x) = (\theta_t, \theta_{ks}, \theta_x)],\end{aligned}$$

holds. In the ideal world, we obtain

$$\Pr[\mathsf{T}_{\mathsf{id}} = \theta] = \left(\frac{1}{2^n}\right)^{q'_{ks}} \cdot \left(\frac{1}{2^n}\right)^{q_x} \cdot \left(\frac{1}{2^\tau}\right)^{q_t}, \tag{16}$$

where q'_{ks} is the number of distinct queries of θ_{ks}, and q_x is the number of $X_i^{\#}[j] \in \mathcal{X}_\$$ for $j \neq 0$. Note that $\Pr[\mathsf{T}_\#^v = \theta_v \mid (\mathsf{T}_\#^t, \mathsf{T}_\#^{ks}, \mathsf{T}_\#^x) = (\theta_t, \theta_{ks}, \theta_x)] = 1$ holds since the adversary always obtains $\perp$ in the ideal world.

In the real world, we first obtain

$$\Pr[\mathsf{T}_{\mathsf{re}}^{ks} = \theta_{ks}] = \left(\frac{1}{2^n}\right)^{q'_{ks}} \tag{17}$$

since there are q'_{ks} distinct queries in θ_{ks}. Second,

$$\Pr[\mathsf{T}_{\mathsf{re}}^x = \theta_x \mid \mathsf{T}_{\mathsf{re}}^{ks} = \theta_{ks}] = \left(\frac{1}{2^n}\right)^{q_x} \tag{18}$$

holds. The reasons are as follows. When $j \neq 0$ and $X_i^{\#}[j] \in \mathcal{X}_{\$}$, we necessarily obtain $X_i^{\#}[j] = \mathsf{R}(X_i^{\#}[j-1]) \oplus B_i^{\#}[j]$, where $X_i^{\#}[j-1]$ is a representative node. Due to $\overline{\mathbf{Bad1}}$, there are no collisions between representative nodes in θ_x. Also, due to $\overline{\mathbf{Bad2}}$, there are no collisions between representative nodes in θ_x and inputs of R in θ_{ks}. Thus, each $X_i^{\#}[j] \in \mathcal{X}_{\$}$ is fixed with probability $1/2^n$. Once all $X_i^{\#}[j] \in \mathcal{X}_{\$}$ for $j \neq 0$ are fixed, all other $X_{i'}^{\#'}[j']$ in θ_x are determined with probability 1 because when $X_{i'}^{\#'}[j'] \in \mathcal{X}_{=} \sqcup \mathcal{X}_{\oplus}$, $X_{i'}^{\#'}[j']$ can be determined as $X_i^{\#}[j']$ or $X_i^{\#}[j'] \oplus B_i^{\#}[j'] \oplus B_{i'}^{\#'}[j']$, where $X_i^{\#}[j'] \in \mathcal{X}_{\$}$. Therefore, Eq. (18) holds. Third, we obtain

$$\Pr[\mathsf{T}_{\mathsf{re}}^{t} = \theta_t \mid (\mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_{ks}, \theta_x)] = \left(\frac{1}{2^\tau}\right)^{q_t}. \tag{19}$$

The reasons are as follows. Since all tagging queries (N_i^t, A_i^t, M_i^t) for $i \in [1..q_t]$ are distinct and $\{B_i^t\}_{i\in[1..q_t]}$ is prefix-free, all inputs of R deriving V_i^t, *i.e.* $X_i^t[\ell_i^t]$, are all representative nodes. Under the condition $(\mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_{ks}, \theta_x)$, all $\mathsf{R}(X_i^t[\ell_i^t])$ are not fixed because **Bad1** and **Bad2** do not happen, and the prefix-freeness of $\{B_i^{\#}\}_{\#\in\{t,v\}, i\in[1..q_{\#}]}$ ensures that $\mathsf{R}(X_i^t[\ell_i^t])$ is not fixed by the element of θ_x. Also, $\overline{\mathbf{Bad1}}$ ensures that all $X_i^t[\ell_i^t]$ are distinct. Thus, we obtain Eq. (19).

Last, we evaluate $\Pr[\mathsf{T}_{\mathsf{re}}^{v} = \theta_v \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)]$. Let $\widehat{V}_i$ be the valid tag for the i-th verification query (N_i^v, A_i^v, M_i^v). We then obtain

$$\begin{aligned}
&\Pr[\mathsf{T}_{\mathsf{re}}^{v} = \theta_v \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \\
&= \Pr[\{V_i^v \neq \widehat{V}_i\}_{i\in[1..q_v]} \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \\
&\geq 1 - \sum_{i=1}^{q_v} \Pr[V_i^v = \widehat{V}_i \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \geq 1 - \frac{q_v}{2^\tau}.
\end{aligned} \tag{20}$$

Here, Eq. (20) holds because $\Pr[V_i^v = \widehat{V}_i \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \leq 1/2^\tau$ holds; when there exists $j \in [1..q_t]$ *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_j^t, A_j^t, M_j^t)$, we obtain $\Pr[V_i^v = \widehat{V}_i \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] = 0$ since $\widehat{V}_i = V_j^t$ holds and **Bad3** does not happen. Suppose that there are no tagging queries *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_j^t, A_j^t, M_j^t)$ for $j \in [1..q_t]$, and we focus on $X_i^v[\ell_i^v]$, which is the input of R deriving $\widehat{V}_i$. When $X_i^v[\ell_i^v]$ is a representative node, similar to the analysis of Eq. (19), $\mathsf{R}(X_i^v[\ell_i^v])$ is not fixed even under the condition $(\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)$ since **Bad1** and **Bad2** do not happen, and $\{B_i^{\#}\}_{\#\in\{t,v\}, i\in[1..q_{\#}]}$ is prefix-free. Thus, we obtain $\Pr[V_i^v = \widehat{V}_i \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \leq 1/2^\tau$. When $X_i^v[\ell_i^v]$ is not a representative node, there exists a unique $i' \in [1..q_v]$ *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_{i'}^v, A_{i'}^v, M_{i'}^v)$ and thus $X_i^v[\ell_i^v] = X_{i'}^v[\ell_{i'}^v]$, where $X_{i'}^v[\ell_{i'}^v] \in \mathcal{X}_{\$}$ because the prefix-freeness ensures that there is previous query *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_j^{\#}, A_j^{\#}, M_j^{\#})$ for $\# \in \{t, v\}$ and $j \in [1..q_{\#}]$, and we here assume that there are no tagging queries *s.t.* $(N_i^v, A_i^v, M_i^v) = (N_j^t, A_j^t, M_j^t)$ for $j \in [1..q_t]$. Similar to the above case, we obtain $\Pr[V_i^v = \widehat{V}_i \mid (\mathsf{T}_{\mathsf{re}}^{t}, \mathsf{T}_{\mathsf{re}}^{ks}, \mathsf{T}_{\mathsf{re}}^{x}) = (\theta_t, \theta_{ks}, \theta_x)] \leq 1/2^\tau$ by using the randomness of $\mathsf{R}(X_{i'}^v[\ell_{i'}^v])$.

From Eqs. (17), (18), (19), (20), we obtain

$$\Pr[\mathsf{T}_{\mathsf{re}} = \theta] \geq \left(\frac{1}{2^n}\right)^{q'_{ks}} \cdot \left(\frac{1}{2^n}\right)^{q_x} \cdot \left(\frac{1}{2^\tau}\right)^{q_t} \cdot \left(1 - \frac{q_v}{2^\tau}\right). \tag{21}$$

From Eqs. (16), (21), we obtain

$$\frac{\Pr[\mathsf{T}_{\mathsf{re}} = \theta]}{\Pr[\mathsf{T}_{\mathsf{id}} = \theta]} \geq 1 - \frac{q_v}{2^\tau}. \tag{22}$$

From Eqs. (15), (22), and Lemma 1, we obtain the following lemma.

Lemma 8.

$$\mathbf{Adv}^{\texttt{Ind+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B}) \leq \frac{(\sigma'_t + \sigma'_v)(0.5\sigma'_t + 0.5\sigma'_v + q_{ks})}{2^n} + \frac{2q_v}{2^\tau}$$

Lemma 8 and Eq. (14) give the bound of $\mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$.

From the simulation of the NMR-INT-RUP game by the UF-CMA$^+$ adversary $\mathcal{B}$, we obtain the following bound.

$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}) \leq \frac{(\sigma'_e + \sigma'_v)(\sigma'_e + \sigma'_v + 2(q_e + q_v + \sigma_e + \sigma_d + \sigma_v))}{2^n} + \frac{4q_v}{2^\tau},$$
$$\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM2[R]}}(\mathcal{A}) \leq \frac{(\sigma'_e + \sigma'_v)(\sigma'_e + \sigma'_v + 2(\sigma_e + \sigma_d + \sigma_v))}{2^n} + \frac{4q_v}{2^\tau}.$$

By combining the above equations and Eqs. (12), (13), we conclude the proof of Theorem 2 and obtain the bound for $\mathsf{CCM2}$.

C.1 Proof of Lemma 6

We here show the proof for the reduction of NMR-INT-RUP security of CCM to UF-CMA$^+$ security of $\mathsf{CBC\text{-}MAC}$. The proof for $\mathsf{CCM2}$ is almost the same as that of CCM; thus, we omit it.

We define a new game, Game-1, against CCM. For Game-1 and an adversary $\mathcal{A}_1$ following Game-1, we relax the NMR-INT-RUP game against $\mathsf{CCM[R]}$ by changing the outputs of $\mathsf{CCM}.\mathcal{E}$ and $\mathsf{CCM}.\mathcal{V}$. When the adversary $\mathcal{A}_1$ queries (N^e, A^e, M^e) and (N^v, A^v, C^v, T^v) to them, suppose it obtains (C^e, U^e, T^e) and (U^v, b), respectively, where $U^\# = \mathsf{R}(\mathsf{ecN}(N^\#, 0))$ for $\# \in \{e, v\}$. We trivially obtain $\mathbf{Adv}^{\texttt{NMR-INT-RUP}}_{\mathsf{CCM[R]}}(\mathcal{A}) \leq \mathbf{Adv}^{\texttt{Game-1}}_{\mathsf{CCM[R]}}(\mathcal{A}_1)$ since Game-1 is the same as NMR-INT-RUP game except that Game-1 additionally reveals $U^\#$. Here, the computational resource of $\mathcal{A}_1$ is the same as that of $\mathcal{A}$.

We then prove $\mathbf{Adv}^{\texttt{Game-1}}_{\mathsf{CCM[R]}}(\mathcal{A}_1) \leq \mathbf{Adv}^{\texttt{UF-CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$. For a simulation of $\mathsf{CCM}.\mathcal{E}$ in Game-1, $\mathcal{B}$ can forward the query of $\mathcal{A}_1$, (N^e, A^e, M^e), to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ and obtain V. Then $\mathcal{B}$ queries $(N^e, 0)$, $(N^e, 1)$, ..., (N^e, m^e) to $\mathcal{KS}_{\mathsf{C}}$ and obtains U^e, KS_1, ..., KS_{m^e}, where $m^e = |M^e|_n$. As a value of $\mathsf{CCM}.\mathcal{E}(N^e, A^e, M^e)$, $\mathcal{B}$ can return $(C^e, U^e, T^e) = (M^e \oplus \mathtt{msb}_{|M^e|}(KS_1 \,\|\, \cdots \,\|\, KS_{m^e}), U^e, V \oplus \mathtt{msb}_\tau(U^e))$ to $\mathcal{A}_1$. Thus, q_e queries to $\mathsf{CCM}.\mathcal{E}$ of $\mathcal{A}_1$ costs q_e queries to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ and

$\sigma_e + q_e$ queries to $\mathcal{KS}_{\mathsf{C}}$ of $\mathcal{B}$. Note that the total number of input blocks to $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ equals σ'_e.

Let (N^d, A^d, C^d, T^d) be the unverified decryption query of $\mathcal{A}_1$. For a simulation of $\mathsf{CCM}.u\mathcal{D}$ in Game-1, $\mathcal{B}$ queries $(N^d, 1)$, $\ldots$, (N^d, m^d) to $\mathcal{KS}_{\mathsf{C}}$ and obtains KS_1, $\ldots$, KS_{m^d}, where $m^d = |C^d|_n$, and it returns $M^d = C^d \oplus \mathtt{msb}_{|C^d|}(KS_1 \| \cdots \| KS_{m^d})$. Thus, q_d queries to $\mathsf{CCM}.u\mathcal{D}$ of $\mathcal{A}_1$ costs σ_d queries to $\mathcal{KS}_{\mathsf{C}}$ of $\mathcal{B}$.

For a simulation of $\mathsf{CCM}.\mathcal{V}$ in Game-1, let (N^v, A^v, C^v, T^v) be the query of $\mathcal{A}_1$. Similar to above, $\mathcal{B}$ queries $(N^v, 0)$, $(N^v, 1)$, $\ldots$, (N^v, m^v) to $\mathcal{KS}_{\mathsf{C}}$ and obtains U^v, $KS_1, \ldots, KS_{m^v}$, where $m^v = |C^v|_n$. Then $\mathcal{B}$ queries (N^v, A^v, M^v, V) to $\mathsf{CBC\text{-}MAC}.\mathcal{V}$, where $M^v = C^v \oplus \mathtt{msb}_{|C^v|}(KS_1 \| \cdots \| KS_{m^v})$ and $V = T^v \oplus \mathtt{msb}_\tau(U^v)$ and obtains b, and it can return (U^v, b) to $\mathcal{A}_1$. Thus, q_v queries to $\mathsf{CCM}.\mathcal{V}$ of $\mathcal{A}_1$ costs q_v queries to $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ and $\sigma_v + q_v$ queries to $\mathcal{KS}_{\mathsf{C}}$ of $\mathcal{B}$. Note that the total number of blocks to execute $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ equals σ'_v. This simulation does not invoke $\mathcal{B}$'s forwarding query from $\mathsf{CBC\text{-}MAC}.\mathcal{T}$ to $\mathsf{CBC\text{-}MAC}.\mathcal{V}$ since Game-1 prohibits the forwarding query from $\mathsf{CCM}.\mathcal{E}$ to $\mathsf{CCM}.\mathcal{V}$.

Therefore, the queries in Game-1 can be simulated by $\mathcal{B}$, and when $\mathcal{A}_1$ obtains $\top$ from $\mathsf{CCM}.\mathcal{V}$, $\mathcal{B}$ also obtains $\top$ from $\mathsf{CBC\text{-}MAC}.\mathcal{V}$. Thus, we obtain $\mathbf{Adv}^{\mathtt{Game\text{-}1}}_{\mathsf{CCM[R]}}(\mathcal{A}_1) \le \mathbf{Adv}^{\mathtt{UF\text{-}CMA+}}_{\mathsf{CBC\text{-}MAC[R]}}(\mathcal{B})$, and this concludes the proof of Lemma 6.

D Proof of Theorem 6

We first focus on CCM and apply PRP/PRF switching lemma.

$$\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM}[E]}(\mathcal{A}) \le \mathbf{Adv}^{\mathtt{PRP}}_{E}(\hat{\mathcal{A}}) + \frac{\sigma_{\mathrm{all}}^2}{2^{n+1}} + \mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM[R]}}(\mathcal{A}'), \tag{23}$$

where $\widehat{\mathcal{A}}$ makes σ_{all} queries, and the computational cost of $\mathcal{A}'$ is the same as $\mathcal{A}$. We then define a function $\mathsf{CCM+}.\mathcal{E}[\mathsf{R}]$ by revising the output of $\mathsf{CCM}.\mathcal{E}[\mathsf{R}]$; it takes (N, A, M) as input and outputs (C, U, V), where $U = \mathsf{R}(\mathsf{ecN}(N, 0))$ and $V = T \oplus \mathtt{msb}_\tau(U)$. Since $\mathsf{CCM+}$ just additionally outputs U compared to CCM, we obtain

$$\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM[R]}}(\mathcal{A}') \le \mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM+[R]}}(\mathcal{B}), \tag{24}$$

where the computational cost of $\mathcal{B}$ is the same as $\mathcal{A}'$. We can evaluate $\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM2}[E]}(\mathcal{A})$ in the same manner as the above, and we obtain

$$\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM2}[E]}(\mathcal{A}') \le \mathbf{Adv}^{\mathtt{PRP}}_{E}(\hat{\mathcal{A}}) + \frac{\sigma_{\mathrm{all}}^2}{2^{n+1}} + \mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM+[R]}}(\mathcal{B}). \tag{25}$$

In what follows, we analyze $\mathbf{Adv}^{\mathtt{NML\text{-}Priv}}_{\mathsf{CCM+[R]}}(\mathcal{B})$.

Let $\mathsf{CCM+\$}.\mathcal{E}[\mathsf{R}]$ be a function obtained by replacing $\mathsf{CBC\text{-}MAC}$ of $\mathsf{CCM+}.\mathcal{E}[\mathsf{R}]$ with a random oracle \$. Thus, $\mathsf{CCM+\$}.\mathcal{E}[\mathsf{R}]$ takes (N, A, M) as input and outputs (C, U, V), where C and U are determined in the same manner as $\mathsf{CCM+}.\mathcal{E}$, but

V is chosen from $\{0,1\}^\tau$ at random. We then obtain the following inequations.

$$\begin{aligned}
\mathbf{Adv}^{\texttt{NML-Priv}}_{\mathsf{CCM+[R]}}(\mathcal{B}) &= |\Pr[\mathcal{B}^{\mathsf{CCM+}.\mathcal{E},\mathsf{CCM+}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+}.\mathcal{E}} = 1]| \\
&= |\Pr[\mathcal{B}^{\mathsf{CCM+}.\mathcal{E},\mathsf{CCM+}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+}.\mathcal{E}} = 1] \\
&\quad + \Pr[\mathcal{B}^{\$,\mathsf{CCM+\$}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+\$}.\mathcal{E}} = 1]| \\
&\le |\Pr[\mathcal{B}^{\mathsf{CCM+}.\mathcal{E},\mathsf{CCM+}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+\$}.\mathcal{E}} = 1]| \\
&\quad + |\Pr[\mathcal{B}^{\$,\mathsf{CCM+}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+\$}.\mathcal{E}} = 1])| \\
&\le 2|\Pr[\mathcal{B}^{\mathsf{CCM+}.\mathcal{E},\mathsf{CCM+}.\mathcal{E}} = 1] - \Pr[\mathcal{B}^{\$,\mathsf{CCM+\$}.\mathcal{E}} = 1]| \\
&=: 2\mathbf{Adv}^{\texttt{NML-Priv\$}}_{\mathsf{CCM+[R]}}(\mathcal{B}). \qquad (26)
\end{aligned}$$

We evaluate $\mathbf{Adv}^{\texttt{NML-Priv\$}}_{\mathsf{CCM+[R]}}(\mathcal{B})$ using H-coefficient technique. Let $(N_i^r, A_i^r, M_i^r, C_i^r, U_i^r, V_i^r)$ and $(N_j^m, A_j^m, M_j^m, C_j^m, U_j^m, V_j^m)$ be i-th nonce-respecting query for $i \in [1..q_r]$ and j-th nonce-misusing query for $j \in [1..q_m]$, respectively. For $\# \in \{r, m\}$ and $i \in [1..q_\#]$, let $m_i^\# = |M_i^\#|_n$; thus, $\sigma_{er} = \sum_{i=1}^{q_r} m_i^r$ and $\sigma_{em} = \sum_{i=1}^{q_m} m_i^m$. Let $\mathsf{ec}(N_i^\#, A_i^\#, M_i^\#) = B_i^\# = (B_i^\#[0], \ldots, B_i^\#[\ell_i^\#])$, and $|B_i^\#|_n = \ell_i^\# + 1$; thus, $\sigma'_\# = \sum_{i=1}^{q_\#}(\ell_i^\# + 1)$.

Similarly to the NMR-INT-RUP proof of $\mathsf{CCM/CCM2}$, we suppose that $\mathcal{B}$ obtains all the block cipher inputs in nonce-respecting and nonce-misusing queries after all queries but before determining an output bit. For $\# \in \{r, m\}$, $i \in [q_\#]$ and $j \in [0..\ell_i^\#]$, let $X_i^\#[j]$ be $j+1$-th block cipher input of $\mathsf{CBC\text{-}MAC}$ in i-th nonce-respecting/nonce-misusing query. Here, we list all queries as $(N_1^r, A_1^r, M_1^r, C_1^r, U_1^r, V_1^r)$, …, $(N_{q_r}^r, A_{q_r}^r, M_{q_r}^r, C_{q_r}^r, U_{q_r}^r, V_{q_r}^r)$, $(N_1^m, A_1^m, M_1^m, C_1^m, U_1^m, V_1^m)$, …, $(N_{q_m}^m, A_{q_m}^m, M_{q_m}^m, C_{q_m}^m, U_{q_m}^m, V_{q_m}^m)$. In the real world, $\mathcal{B}$ obtains the real values of $X_i^\#[j]$. In the ideal world, $\mathcal{B}$ obtains the dummy values of $X_i^\#[j]$, which are determined in the same way as in the NMR-INT-RUP proof of CCM.

Including all the revealed $X_i^\#[j]$, we define a transcript $\theta = (\theta_r, \theta_m, \theta_x)$ of $\mathcal{B}$ such that

- $\theta_r = \{(N_i^r, A_i^r, M_i^r, C_i^r, U_i^r, V_i^r)\}_{i \in [1..q_r]}$,
- $\theta_m = \{(N_i^m, A_i^m, M_i^m, C_i^r, U_i^r, V_i^m)\}_{i \in [1..q_m]}$,
- $\theta_x = \{X_i^\#[j]\}_{\# \in \{r,m\}, i \in [1..q_\#], j \in [0..\ell_i^\#]}$.

Bad event evaluation. We define $\theta \in \Theta_{\mathrm{bad}}$ if at least one of the following events happens in θ.

Bad1 Collision of $X[\cdot]$ corresponding to representative nodes in θ_x: there exists $\#_1, \#_2 \in \{r, m\}$, $i_1 \in [1..q_{\#_1}]$, $i_2 \in [1..q_{\#_2}]$, $j_1 \in [0..\ell_{i_1}^{\#_1}]$, $j_2 \in [0..\ell_{i_2}^{\#_2}]$ *s.t.* $X_{i_1}^{\#_1}[j_1], X_{i_2}^{\#_2}[j_2] \in \mathcal{X}_\$ \sqcup \mathcal{X}_\oplus$ and $X_{i_1}^{\#_1}[j_1] = X_{i_2}^{\#_2}[j_2]$.

Bad2 Collision of $X[\cdot]$ corresponding to representative nodes in θ_x and $\mathsf{ecN}(N^r, \cdot)/\mathsf{ecN}(N^m, \cdot)$ in θ_r/θ_m: there exists $\#_1, \#_2 \in \{r, m\}$, $i_1 \in [1..q_{\#_1}]$, $j_1 \in [0..\ell_{i_1}^{\#_1}]$, $i_2 \in [1..q_{\#_2}]$, $j_2 \in [0..m_{i_2}^{\#_2}]$ *s.t.* $X_{i_1}^{\#_1}[j_1] \in \mathcal{X}_\$ \sqcup \mathcal{X}_\oplus$ and $X_{i_1}^{\#_1}[j_1] = \mathsf{ecN}(N_{i_2}^{\#_2}, j_2)$.

We can evaluate the upper bound of $\Pr[\mathsf{T}_{\mathsf{id}} \in \Theta_{\mathrm{bad}}] \coloneqq \Pr[\mathbf{Bad1} \cup \mathbf{Bad2}]$ in the same manner as that in the NMR-INT-RUP proof of CCM; thus, we obtain

$$\begin{aligned}\Pr[\mathsf{T}_{\mathsf{id}} \in \Theta_{\mathrm{bad}}] &\leq \frac{(\sigma'_r + \sigma'_m)^2}{2^{n+1}} + \frac{(\sigma'_r + \sigma'_m)(\sigma_{er} + \sigma_{em} + q_r + q_m)}{2^n} \\ &= \frac{(\sigma'_r + \sigma'_m)(0.5\sigma'_r + 0.5\sigma'_m + \sigma_{er} + \sigma_{em} + q_r + q_m)}{2^n}. \quad (27)\end{aligned}$$

Good transcript ratio. For $i \in [1..q_m]$, we split transcript $\theta_m = \{(N_i^m, A_i^m, M_i^m, C_i^r, U_i^r, V_i^m)\}$ into $\theta_{m1} = \{(N_i^m, A_i^m, M_i^m, C_i^r, U_i^r)\}$ and $\theta_{m2} = \{V_i^m\}$. For $\# \in \{\mathsf{re}, \mathsf{id}\}$, let $\mathsf{T}_\#^r$, $\mathsf{T}_\#^{m1}$, $\mathsf{T}_\#^{m2}$, $\mathsf{T}_\#^x$ denote the random variables of θ_r, θ_{m1}, θ_{m2}, θ_x in each world, respectively. For a good transcript $\theta \in \Theta_{\mathrm{good}}$, we have

$$\begin{aligned}\Pr[\mathsf{T}_\# = \theta] = \Pr[\mathsf{T}_\#^{m1} = \theta_{m1}] &\cdot \Pr[\mathsf{T}_\#^x = \theta_x \mid \mathsf{T}_\#^{m1} = \theta_{m1}] \\ &\cdot \Pr[\mathsf{T}_\#^r = \theta_r \mid (\mathsf{T}_\#^{m1}, \mathsf{T}_\#^x) = (\theta_{m1}, \theta_x)] \\ &\cdot \Pr[\mathsf{T}_\#^{m2} = \theta_{m2} \mid (\mathsf{T}_\#^r, \mathsf{T}_\#^{m1}, \mathsf{T}_\#^x) = (\theta_r, \theta_{m1}, \theta_x)].\end{aligned}$$

We first obtain $\Pr[\mathsf{T}_{\mathsf{re}}^{m1} = \theta_{m1}] = \Pr[\mathsf{T}_{\mathsf{id}}^{m1} = \theta_{m1}]$ because we define $\mathsf{CCM}{+}\$.\mathcal{E}$ to output (C, U) in the same distribution as $\mathsf{CCM}{+}.\mathcal{E}$. In both worlds, we obtain

$$\Pr[\mathsf{T}_\#^x = \theta_x \mid \mathsf{T}_\#^{m1} = \theta_{m1}] = \left(\frac{1}{2^n}\right)^{q_x}, \quad (28)$$

$$\Pr[\mathsf{T}_\#^r = \theta_r \mid (\mathsf{T}_\#^{m1}, \mathsf{T}_\#^x) = (\theta_{m1}, \theta_x)] = \left(\frac{1}{2}\right)^{\sum_{i=1}^{q_r} |M_i^r|} \left(\frac{1}{2^n}\right)^{q_r} \left(\frac{1}{2^\tau}\right)^{q_r}, \quad (29)$$

$$\Pr[\mathsf{T}_\#^{m2} = \theta_{m2} \mid (\mathsf{T}_\#^r, \mathsf{T}_\#^{m1}, \mathsf{T}_\#^x) = (\theta_r, \theta_{m1}, \theta_x)] = \left(\frac{1}{2^\tau}\right)^{q_m}, \quad (30)$$

where q_x is the number of $X_i^\#[j] \in \mathcal{X}_\$$ for $j \neq 0$. In the ideal world, Eq. (28) holds since all $X_i^\#[j] \in \mathcal{X}_\$$ for $j \neq 0$ are randomly chosen, and Eqs.(29) and (30) hold since (C_i^r, U_i^r, V_i^r) and V_j^m for $i \in [1..q_r]$, $j \in [1..q_m]$ are output from the random oracles.

In the real world, Eq. (28) holds in the same manner as Eq. (18). All $X_i^\#[j] \in \mathcal{X}_\$$, where $j \neq 0$ can be derived using representative node $X_i^\#[j-1]$; *i.e.*, $X_i^\#[j] = \mathsf{R}(X_i^\#[j-1]) \oplus B_i^\#[j]$. $\overline{\mathbf{Bad1}}$ and $\overline{\mathbf{Bad2}}$ enables us to fix all $X_i^\#[j] \in \mathcal{X}_\$$, where $j \neq 0$, with the probability $(1/2^n)^{q_x}$. Moreover, once all $X_i^\#[j] \in \mathcal{X}_\$$ for $j \neq 0$ are fixed, all other $X_i^\#[j]$ is fixed with probability 1. Regarding Eq. (29), $\overline{\mathbf{Bad2}}$ and $\mathcal{N}_1 \cap \mathcal{N}_2 = \emptyset$ enables us to fix C_i^r and U_i^r with probability $(1/2)^{\sum_{i=1}^{q_r} |M_i^r|}$ and $(1/2^n)^{q_r}$, respectively. Also, V_i^r can be fixed in the same manner as Eq. (19); distinctness of queries and prefix-freeness ensure all $X_i^r[\ell_i^r]$, which are the input of R deriving V_i^r, are representative nodes. $\overline{\mathbf{Bad1}}$ ensures that all $X_i^r[\ell_i^r]$ are distinct, and $\mathsf{R}(X_i^r[\ell_i^r])$ is not fixed due to $\overline{\mathbf{Bad1}}$, $\overline{\mathbf{Bad2}}$, and prefix-freeness. Similarly, we obtain Eq. (30).

From Eqs. (28), (29), (30), and $\Pr[\mathsf{T}_{\mathsf{re}}^{m1} = \theta_{m1}] = \Pr[\mathsf{T}_{\mathsf{id}}^{m1} = \theta_{m1}]$, we obtain

$$\frac{\Pr[\mathsf{T}_{\mathsf{re}} = \theta]}{\Pr[\mathsf{T}_{\mathsf{id}} = \theta]} = 1 \quad (31)$$

Algorithm $\Theta\mathsf{CB3}.\mathcal{E}[\widetilde{E}_K](N, A, M)$

$\Sigma \leftarrow 0^n$
$(M[1], \ldots, M[m-1]) \overset{n}{\leftarrow} M$
for $i = 1$ **to** $m - 1$ **do**
  $C[i] \leftarrow \widetilde{E}_K^{(N,i)}(M[i])$
  $\Sigma \leftarrow \Sigma \oplus M[i]$
if $|M[m]| = n$ **then**
  $C[m] \leftarrow \widetilde{E}_K^{(N,m)}(M[m])$
  $\Sigma \leftarrow \Sigma \oplus M[m]$
  $V \leftarrow \widetilde{E}_K^{(N,m,\$)}(\Sigma)$
else
  $C[m] = \mathtt{msb}_{|M[m]|}(\widetilde{E}_K^{(N,m,*)}(0^n) \oplus M[m])$
  $\Sigma \leftarrow \Sigma \oplus (M[m] \,\|\, 10^*)$
  $V \leftarrow \widetilde{E}_K^{(N,m,*\$)}(\Sigma)$
$T \leftarrow \mathtt{msb}_\tau(V \oplus \mathrm{Hash}[\widetilde{E}_K](A))$
$C \leftarrow C[1] \,\|\, \cdots \,\|\, C[m-1] \,\|\, C[m]$
return (C, T)

Algorithm $\mathrm{Hash}[\widetilde{E}_K](N, A, M)$

$\mathit{\Gamma} \leftarrow 0^n$
$(A[1], \ldots, A[a-1]) \overset{n}{\leftarrow} A$
for $i = 1$ **to** $a - 1$ **do**
  $\mathit{\Gamma} \leftarrow \mathit{\Gamma} \oplus \widetilde{E}_K^{(i)}(A[i])$
if $|A[a]| = n$ **then**
  $\mathit{\Gamma} \leftarrow \mathit{\Gamma} \oplus \widetilde{E}_K^{(m)}(A[m])$
else
  $\mathit{\Gamma} \leftarrow \mathit{\Gamma} \oplus \widetilde{E}_K^{(m,*)}(A[m] \,\|\, 10^*)$
return $\mathit{\Gamma}$

Algorithm $\Theta\mathsf{CB3}.\mathcal{D}[\widetilde{E}_K](N, A, C, T)$

$\Sigma \leftarrow 0^n$
$(C[1], \ldots, C[m-1]) \overset{n}{\leftarrow} C$
for $i = 1$ **to** $m - 1$ **do**
  $M[i] \leftarrow \widetilde{D}_K^{(N,i)}(C[i])$
  $\Sigma \leftarrow \Sigma \oplus M[m]$
if $|C[m]| = n$ **then**
  $M[m] \leftarrow \widetilde{D}_K^{(N,m)}(C[m])$
  $\Sigma \leftarrow \Sigma \oplus M[m]$
  $V \leftarrow \widetilde{E}_K^{(N,m,\$)}(\Sigma)$
else
  $M[m] = \mathtt{msb}_{|C[m]|}(\widetilde{D}_K^{(N,m,*)}(0^n) \oplus C[m])$
  $\Sigma \leftarrow \Sigma \oplus (M[m] \,\|\, 10^*)$
  $V \leftarrow \widetilde{E}_K^{(N,m,*\$)}(\Sigma)$
$\widehat{T} \leftarrow \mathtt{msb}_\tau(V \oplus \mathrm{Hash}[\widetilde{E}_K](A))$
if $T \neq \widehat{T}$ **then return** $\perp$
else
  $M \leftarrow M[1] \,\|\, \cdots \,\|\, M[m-1] \,\|\, M[m]$
  return M

Fig. 9: Algorithms of $\Theta\mathsf{CB3}[\widetilde{E}_K]$.

for $\theta \in \mathit{\Theta}_{\mathrm{good}}$. From Eqs. (27) and (31), we obtain the following bound.

$$\mathbf{Adv}_{\mathsf{CCM+[R]}}^{\mathtt{NML\text{-}Priv\$}}(\mathcal{B}) \leq \frac{(\sigma'_r + \sigma'_m)(0.5\sigma'_r + 0.5\sigma'_m + \sigma_{er} + \sigma_{em} + q_r + q_m)}{2^n}. \quad (32)$$

Combining Eqs. (23), (24), (26), (32) and combining Eqs. (25), (26), (32) give the NML-Priv bounds for CCM and CCM2, respectively.

E Algorithms of ΘCB3

Figure 9 shows the algorithms of ΘCB3.